Mutator Party: Product, Design and Build Spec

Version: v3 plan (documents the working v2 demo and everything we decided to add) **Date:** 11 October 2026 **Status:** Planning document. The demo in section 2 is real and playable. Everything marked **MOCKUP** or **PLANNED** is not built yet.

> **Read this first.** Sections 1 to 3 explain the idea and what already works. Sections 4 to 7 are the new features you asked for (name prompt, host controls, random arenas, first-to-reach power-ups, themes). Section 8 lists every font, color and icon choice. Section 10 is how to make money. Section 11 is the long list of things that are easy to forget. Section 13 has copy-paste-ready code.

Table of contents

1. The idea in one page
2. What exists today (the v2 demo)
3. Player journey and the name prompt
4. Host controls: the Create Room screen
5. Randomized arenas (a new layout every round)
6. Power-ups and the "first to reach" rule
7. Themes and round-wise theme change
8. Design system: fonts, colors, icons, sound, motion
9. Technical architecture
10. How to make money
11. Things you may have forgotten (care checklist)
12. Roadmap and effort
13. Appendices (settings schema, twist catalog, physics constants, protocol, code, test plan)

1. The idea in one page

Pitch: a free, instant-play party game in the browser. 2 to 12 friends join with one link. Every round is a different **game mode** plus one to three **rule twists** (low gravity, reversed controls, crumbling floor and so on). Rounds last about a minute, so a full session is about 5 to 8 minutes.

Why this can spread (lessons from Skribbl.io and Smash Karts, your two favourites):

What those games do	What we copy
No login. You are playing in about 10 seconds.	Name prompt only, no account.
The rules fit in one sentence.	"Shove everyone off." "Don't hold the bomb." "Hold the crown."
Short rounds, so "one more game" is easy.	45 to 75 second rounds.
Players create the funny moments, not the designer.	Chaos twists plus power-ups plus player-picked rules.
Friends invite friends with a link.	Room codes and a WhatsApp share button.
Cheap to run.	Small 2D game, tiny network messages.

Five product pillars

1. **Instant.** Link to playing in under 15 seconds.
2. **Chaos you can explain in one sentence.**
3. **Host control.** The room creator decides the rules, so every room feels like "our game".
4. **Fair comeback.** The player in last place picks the next twist.
5. **Clip-able.** Every big moment (a knockout, a bomb blast, a crown steal) is worth a screenshot or a short clip.

Glossary

Word	Meaning
Room	One private match with a code, like `KPQ7X`.
Host	The player who created the room. Controls the rules.
Mode	The objective of a round: Last One Standing, Bomb Tag, Crown Rush.
Twist (also called hurdle or mutator)	A rule change such as Moon Gravity. 1 to 3 per round.
Hazard	A physical obstacle (spikes, bumpers). Planned, not built.
Crate / power-up	A pickup that gives one special move.
Power spot	A fixed location where a crate appears.
Arena	The platforms and island for one round.
Theme	The visual style (colors, background, particles, music).

2. What exists today (the v2 demo)

The v2 demo is one HTML file. You play against 7 bots on one screen. Nothing is online yet.

2.1 Screenshots (real, captured from the game code)

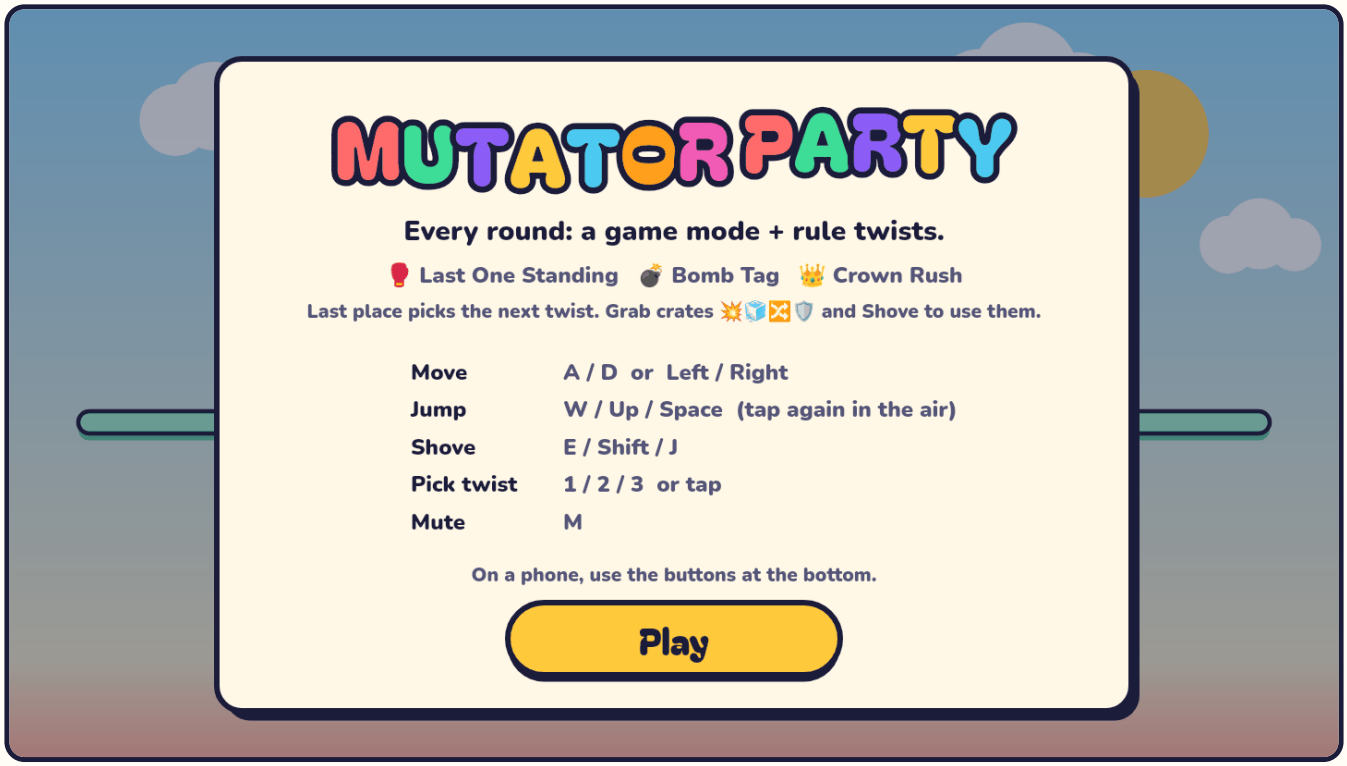


Figure 2.1: Menu. Controls and the three modes are listed.

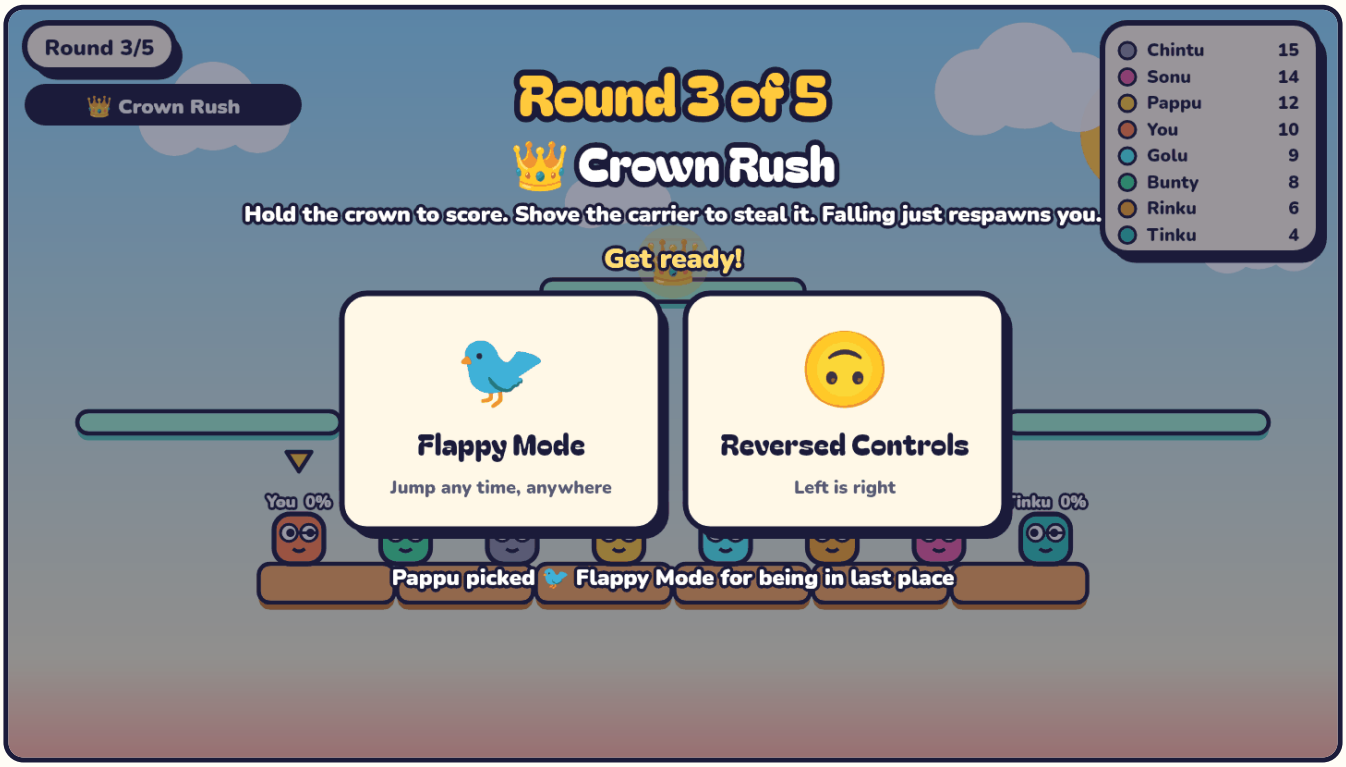


Figure 2.2: Round intro. The mode is shown first, then the twist cards lock in one by one. The last-place player (Pappu) picked the first twist.

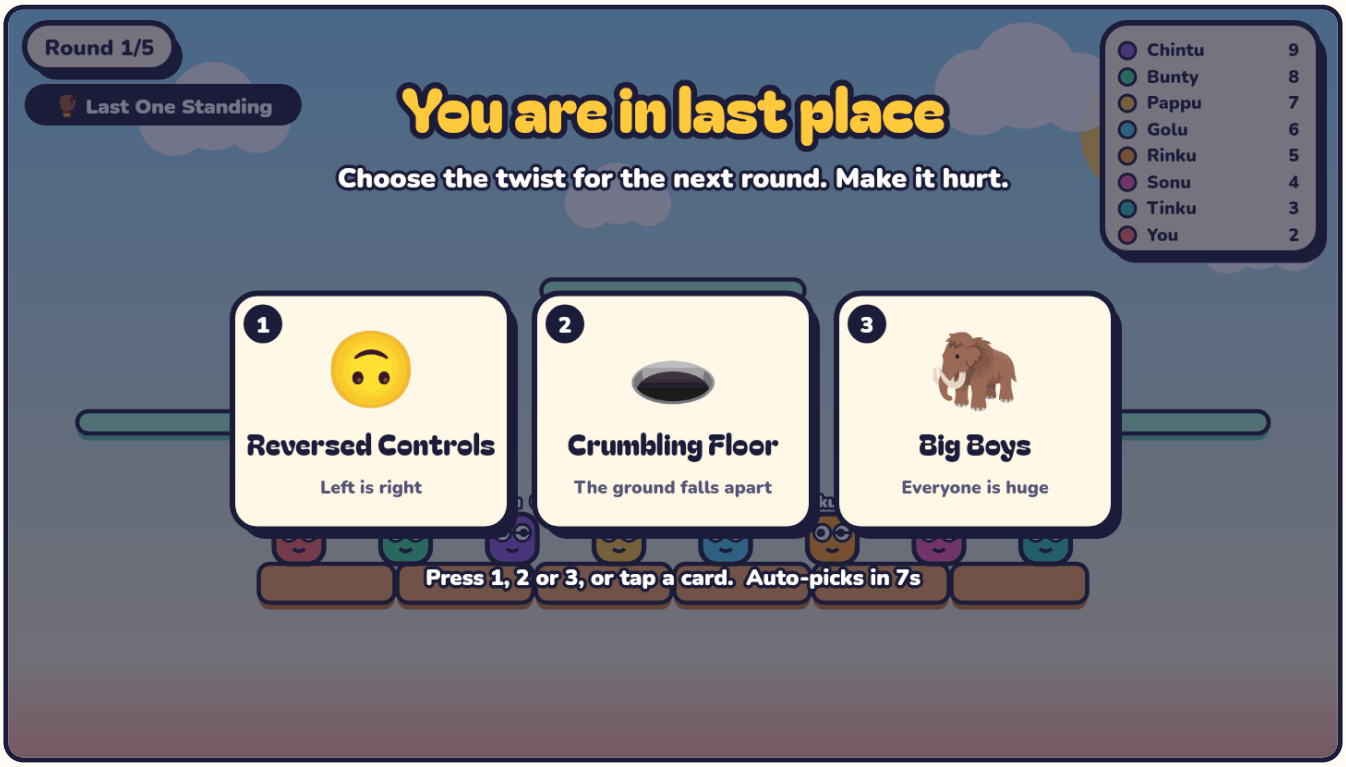


Figure 2.3: When you are in last place you choose between 3 twist cards (keys 1, 2, 3 or tap). This is the "selecting option" you liked.



Figure 2.4: Last One Standing with the Size Lottery twist. A power-up crate sits on the island.

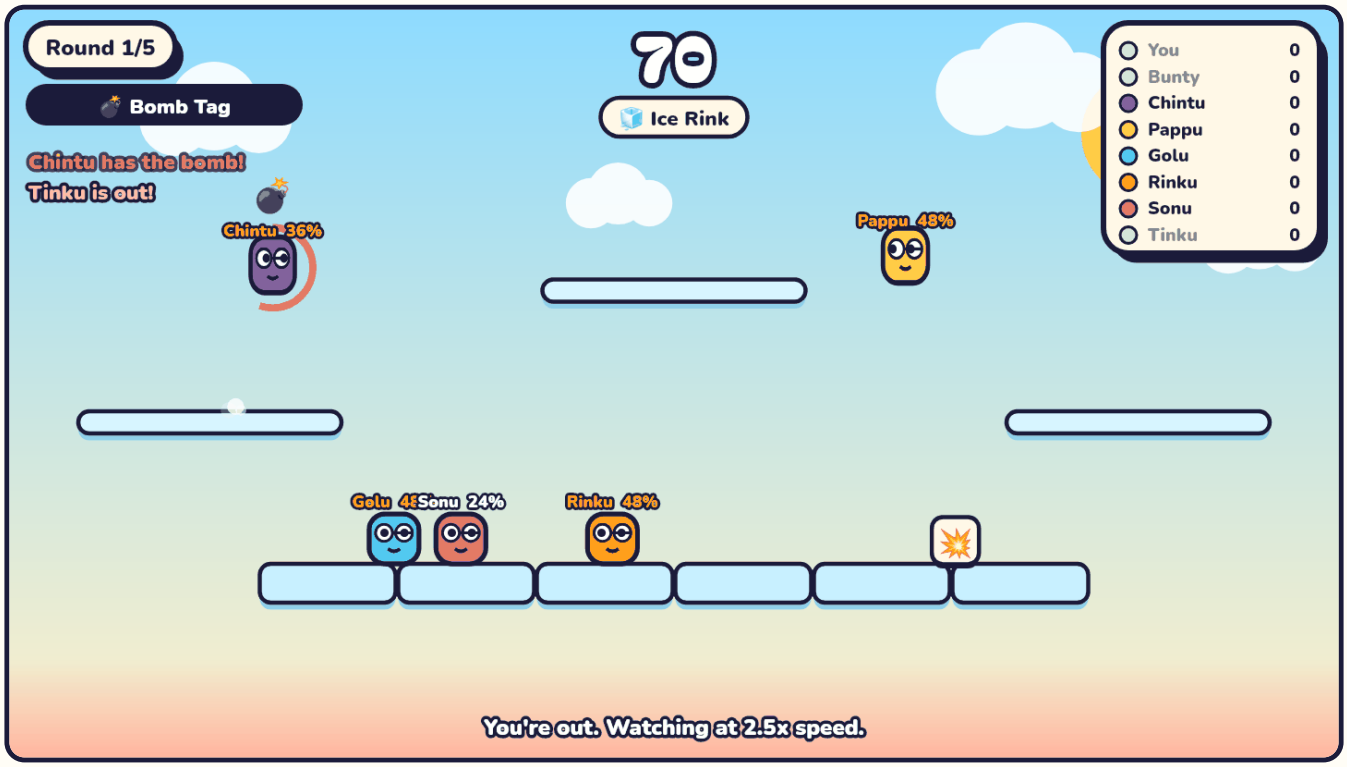


Figure 2.5: Bomb Tag with Ice Rink. The holder has a bomb icon and a red fuse ring. The feed on the left narrates events.

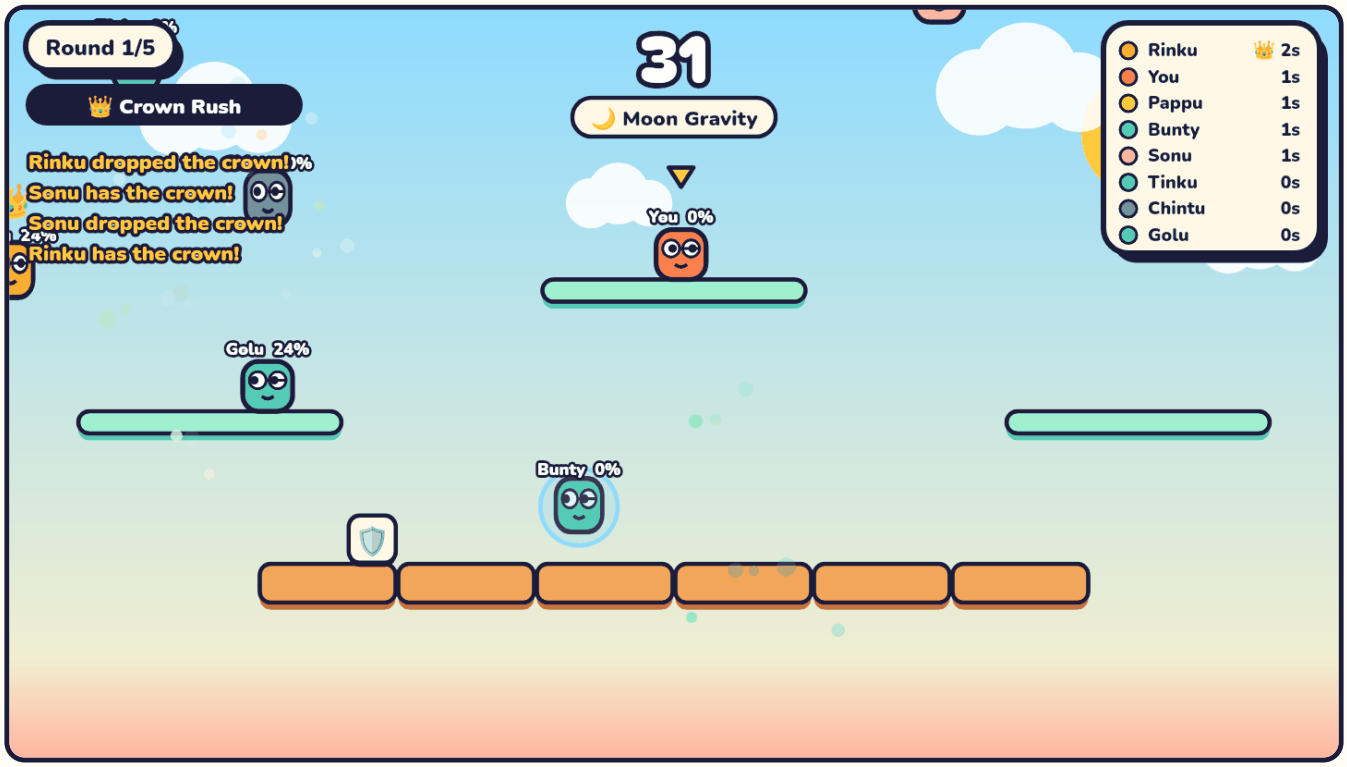


Figure 2.6: Crown Rush with Moon Gravity. The scoreboard switches to crown seconds. A shield crate is on the floor.

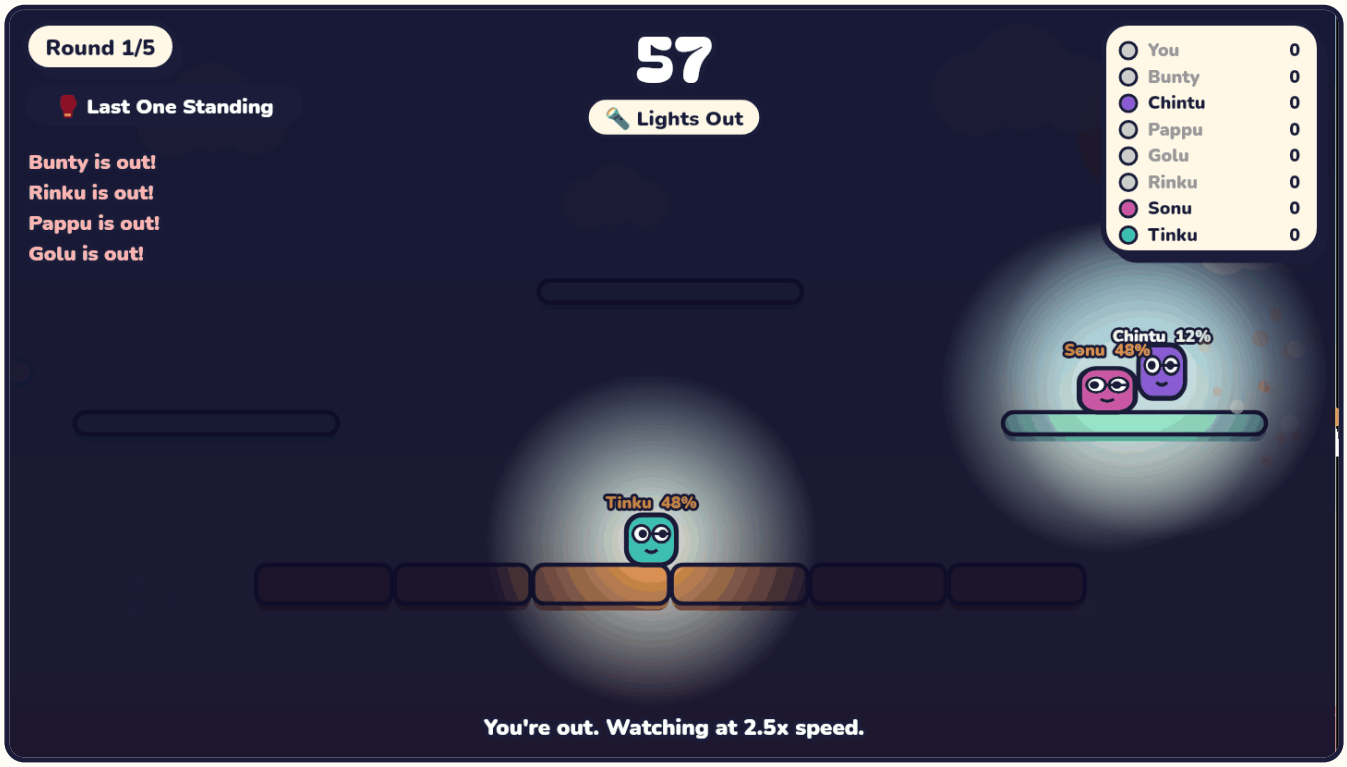


Figure 2.7: Lights Out. You only see players near a light circle.

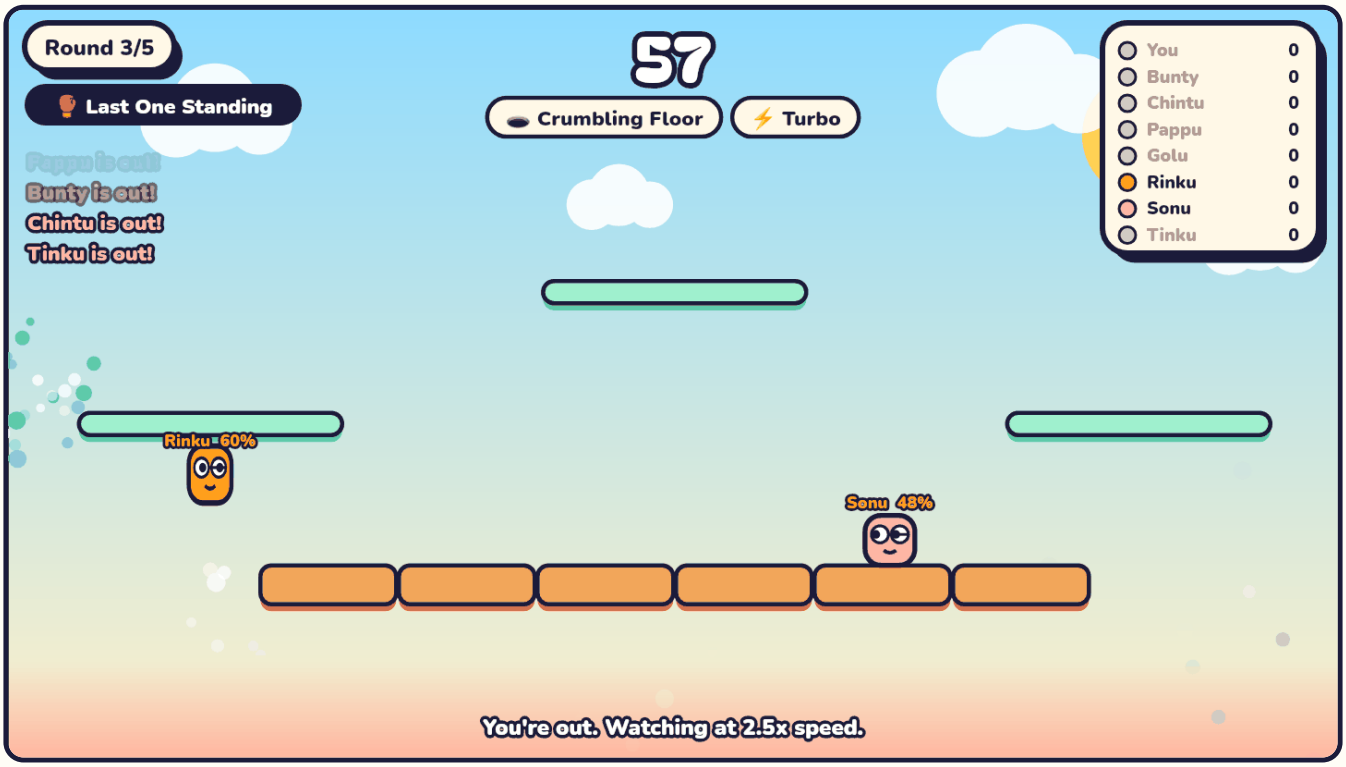


Figure 2.8: Crumbling Floor plus Turbo. The red tile is about to fall.

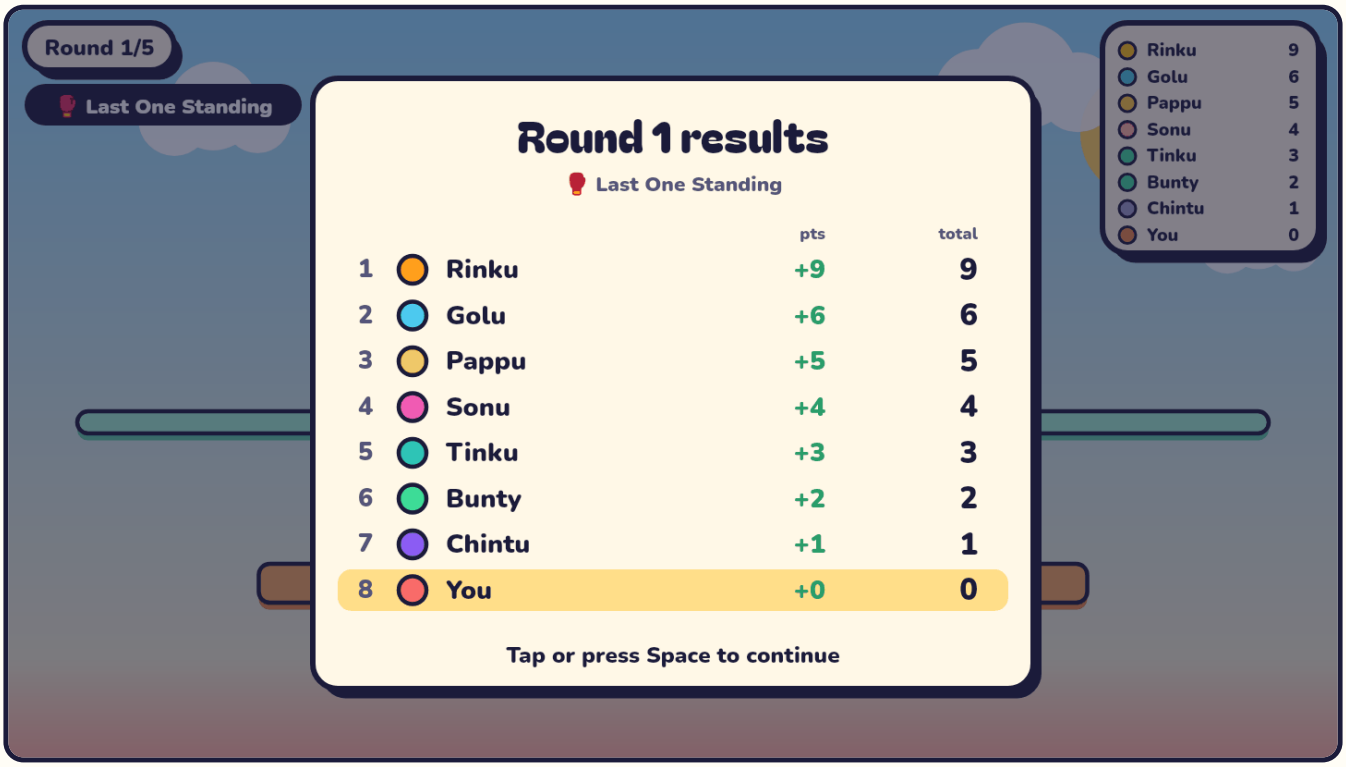


Figure 2.9: Round results with points and totals.



Figure 2.10: Final standings after 5 rounds.

2.2 What is in the demo

- 3 modes: Last One Standing, Bomb Tag, Crown Rush.
- 13 twists in 5 categories (gravity, size, controls, jumping, world).
- 4 power-ups: Mega Shove, Freeze Burst, Swap, Shield.
- "Last place picks the next twist" with 3 cards.
- 7 bots (player vs bots on one screen), keyboard and touch controls.
- Game feel: freeze-frame on hard hits, slow motion on the last knockout, screen shake, text feed.

2.3 Honest list of what is missing or wrong

Problem	Why it matters
No online play.	The real product needs rooms, a server and many humans.
Arena is the same every round.	You noticed this. Fixed in section 5.
Rounds are much too short in bot-only tests.	See 2.4.
Bots are simple (chase, flee, shove).	Fine for a demo, not for fun solo play.
No name entry, no host settings, no themes.	Sections 3, 4 and 7.
Keys use the typed letter, not the key position.	Breaks on non-QWERTY keyboards. Use `event.code` (section 11).
Emoji icons look different on every phone.	Section 8.3 has the fix.

2.4 Measured balance problem (important)

While preparing these docs I ran the game code headless with 8 bots (the human player idle) and timed how long each round lasted:

Mode	Rounds simulated	Average	Shortest	Median	Longest	Intended
Last One Standing	24	11.8 s	3.9 s	10.0 s	45.5 s	40 to 60 s
Bomb Tag	24	9.8 s	3.9 s	10.8 s	18.2 s	45 to 75 s
Crown Rush	8	46.5 s	46.5 s	46.5 s	46.5 s	45 s (timer)

Brawl and Bomb rounds end four to six times faster than intended. The likely causes (not yet fixed, listed so we can tune them):

1. Shove knockback is too strong: base force 620 plus 70 per hit is large for a 600 px island.
2. Bots walk off edges and chase targets near edges (the "suicidal bot" chance is 8 percent per decision).
3. Bomb Tag passes also kick both players sideways (260 px/s) and the explosion shockwave (760 px/s) pushes bystanders off the island.
4. There is no safety margin at the island edge and no ring-out grace time.

Tuning targets for v3: Brawl 40 to 60 s, Bomb 45 to 75 s. Start by lowering base knockback to about 420, removing the sideways kick on bomb passes, adding an edge-safety check for bots, and capping shockwave speed at about 500. Re-run the same timing test after every change (section 13.8).

3. Player journey and the name prompt

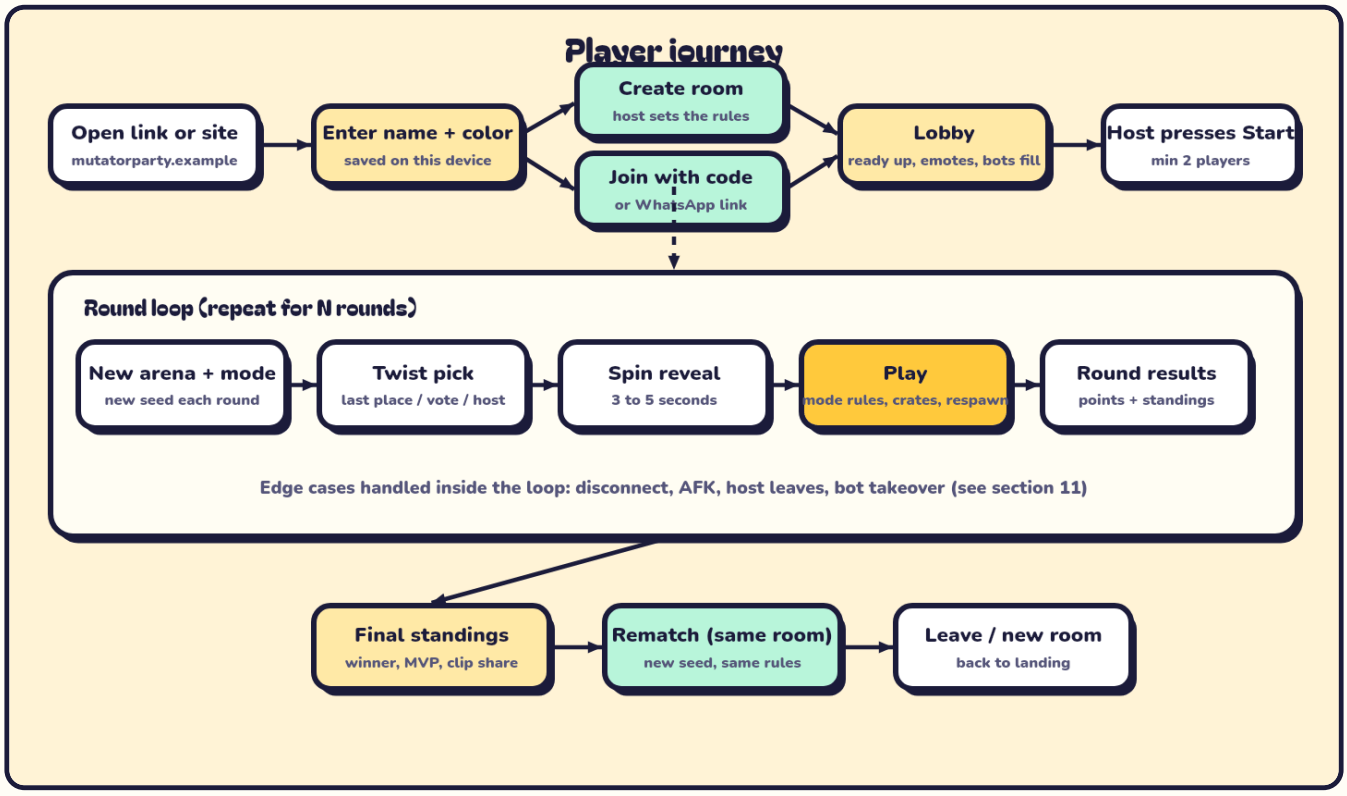


Figure 3.1: Player journey from first visit to rematch.

3.1 Name prompt (your first requirement)

When it appears: the first time someone opens the game (home page, a room link, or a room code). The prompt must appear **before** they can create or join a room.



Figure 3.2 (MOCKUP): Landing screen. 1 = name box with a dice button for a random name. 2 = color picker. 3 = Create a room. 4 = Join with code.

Rules for the name

Rule	Value	Reason
Length	2 to 16 characters, trimmed	Fits on screen and in the scoreboard.
Allowed	Letters (any language), digits, space, `_`, `-`, `'`	Blocks symbols used for tricks.
Not allowed	URLs, e-mails, phone numbers, repeated spaces	Privacy and spam.
Reserved	`You`, `Host`, `Admin`, `Bot`, `System`, names of the bots	Prevents impersonation.
Profanity	Filter on the server (English and Hinglish list, plus leetspeak and look-alike letters)	The server must check, never only the browser.
Duplicates	If taken in the room, add `#2`, `#3`	Everyone stays distinguishable.
Random name	Dice button gives something like "Sleepy Samosa"	Removes the blank-page problem, great for mobile.
Memory	Save the name and color in the browser (`localStorage`)	Returning players skip the prompt (but can edit).
Privacy hint	"Please do not use your real full name."	Many players are young.

Guest identity: at first visit the browser creates a random `playerId` (UUID) and the server gives back a signed `sessionToken`. This allows reconnecting after a dropped connection without any account.

Joining by link: `https://yourdomain/r/KPQ7X`. If the player has no saved name, show the name prompt first and then drop them into the lobby of room `KPQ7X`.

3.2 Lobby

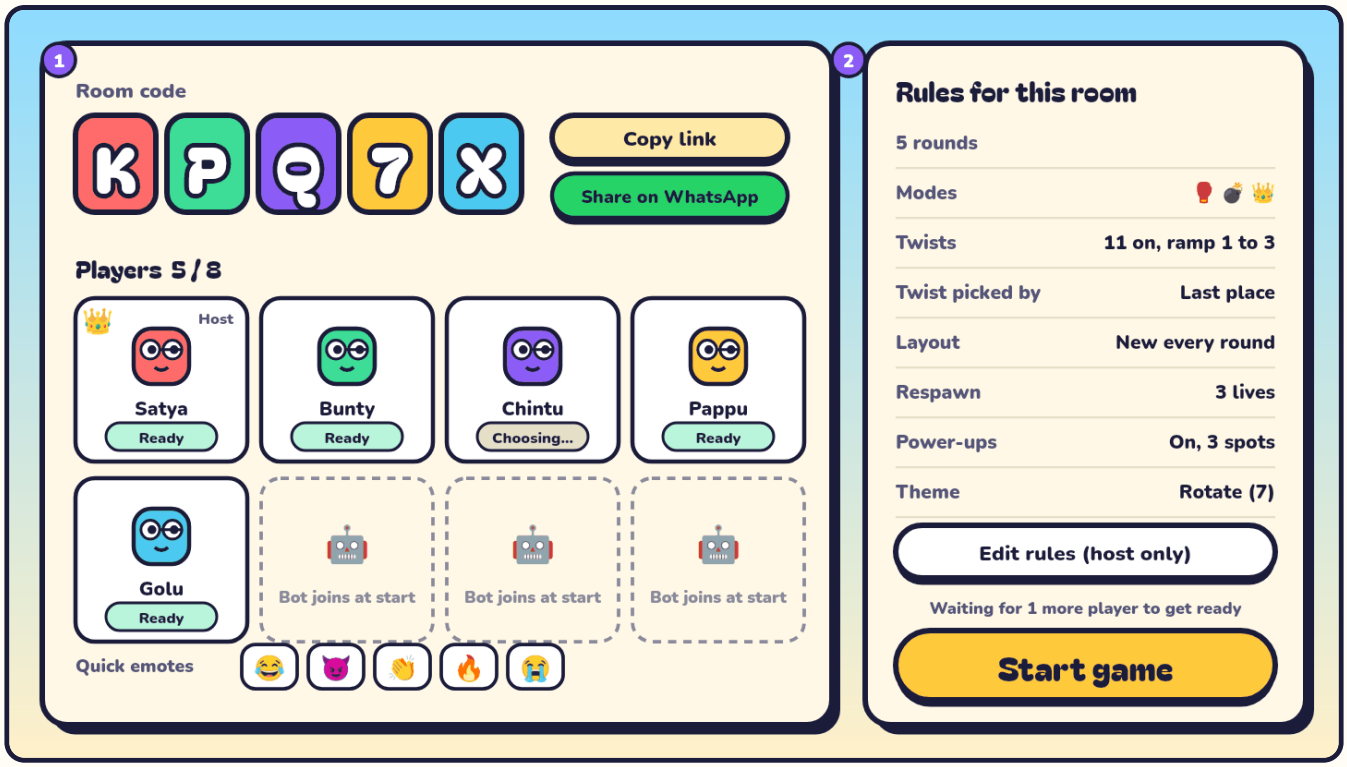


Figure 3.3 (MOCKUP): Lobby. 1 = room code and share buttons. 2 = rules summary. The host sees Edit rules and Start.

Element	Behaviour
Room code	5 characters from an alphabet without look-alikes (no 0, O, 1, I, L). Big, tap to copy.
Share	"Copy link" and "Share on WhatsApp" (use the Web Share API on phones).
Players	Shows up to max players. Each shows Ready or Choosing. Host has a crown.
Bots	Empty slots fill with bots at start if the host allows it.
Rules summary	Read-only for guests, editable for the host.
Start	Host only. Needs at least 2 players (bots count if allowed).
Emotes	A small fixed set of emoji (safe by design, no free text needed).

3.3 Edge cases to design for

Situation	What should happen
Player loses connection	Keep their slot for 30 seconds. A bot plays for them. They resume when they reconnect.
Host leaves	The longest-connected human becomes host. Announce it in the feed.
Room full	Show "Room full" with option to spectate if spectators are allowed.
Wrong code	"We could not find that room." No hints about similar codes.
Two players pick the same name	Auto-suffix `#2`.
Player idle (AFK)	Warn at 15 s of no input in a round, bot takes over at 30 s, move to spectator after 2 missed rounds.

Situation	What should happen
Game already started	New arrivals watch until the next round or the next match.

4. Host controls: the Create Room screen

This is the screen where the creator chooses **everything**: how many platforms, how many rounds, how many players, which hurdles are allowed, respawn on or off, themes and more. The letters A to G below match the letters on the mockup.

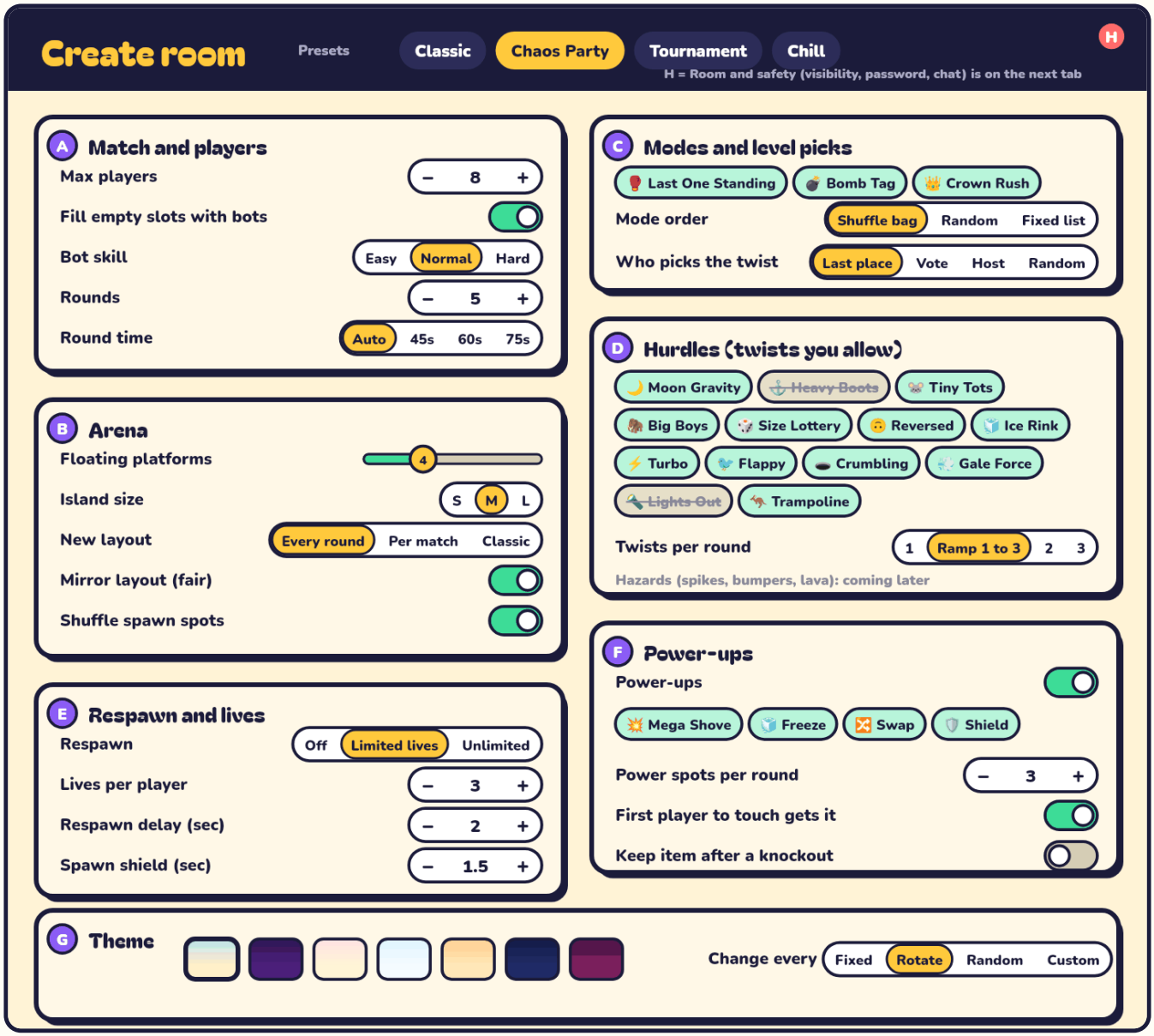


Figure 4.1 (MOCKUP): Create Room. Sections A to G. A section H (room and safety) sits on a second tab.

4.1 Presets (one tap, then tweak)

Preset	What it sets
Classic	3 modes, 1 twist per round, respawn off, fixed layout, no power-ups.
Chaos Party (default)	All modes and twists, ramp 1 to 3 twists, new layout each round, power-ups on, 3 lives.
Tournament	10 rounds, no power-ups, no respawn, mirrored fixed seed, last-place picks off.
Chill	Fewer twists, unlimited respawn, big island, easy bots.
Custom	Anything the host changes.

4.2 Every setting (complete list)

A. Match and players

Setting	Type	Default	Range / options	Notes
`maxPlayers`	number	8	2 to 12	Hard cap 12 for the MVP (performance and screen space).
`minPlayersToStart`	number	2	2 to max	Bots can fill the rest.
`fillBots`	on/off	on		Bots join at start so rooms never feel empty.
`botSkill`	choice	normal	easy, normal, hard	Changes reaction time and edge awareness.
`rounds`	number	5	1 to 15	Also offer "Best of" later.
`roundTime`	choice	auto	auto, 30 to 120 s	Auto uses the mode default (60 / 75 / 45).
`scoring`	choice	standard	standard, winner_plus, kills	Standard = placement points, bonus for a clear winner.
`spectators`	on/off	on		People who join late or are out.

B. Arena (see section 5)

Setting	Type	Default	Range / options	Notes
`platformCount`	number	4	2 to 8	Floating platforms only. The island is separate.
`islandSize`	choice	M	S (4 tiles), M (6), L (8)	Tile width is 100 px.
`layout`	choice	every_round	every_round, per_match, classic	Classic = the fixed v2 layout.
`mirror`	on/off	on		Left-right symmetric layout. Fairer.
`shuffleSpawns`	on/off	on		Who starts where changes every round.

C. Modes and level-wise picks

Setting	Type	Default	Options	Notes
`modes.pool`	multi-select	all 3	brawl, bomb, crown (more later)	Host can exclude modes.

Setting	Type	Default	Options	Notes
`modes.order`	choice	shuffle_bag	shuffle_bag, random, fixed	Shuffle bag = every mode once before repeating. Fixed = host orders them round by round.
`modes.twistPicker`	choice	last_place	last_place, vote, host, random, round_robin	This is the "random level-wise pick" family.

How each twist picker works:

Picker	Flow
last_place	Lowest total score sees 3 random twist cards and picks one. A bot picks if it is a bot.
vote	Everyone sees 3 cards for 8 seconds. Most votes wins, ties are random.
host	The host picks every round. Good for tournaments.
random	The server picks.
round_robin	Each player picks once in turn.

D. Hurdles (twists and hazards the host allows)

Setting	Type	Default	Notes
`twists.allowed`	multi-select	all 13	Tick or untick each one. Empty list is allowed (no twists).
`twists.perRound`	choice	ramp	1, 2, 3 or ramp (1, 1, 2, 2, 3).
`twists.themeLinked`	on/off	off	If on, the theme brings its own twist (Frozen Peak adds Ice Rink).
`hazards.allowed`	multi-select	none	Planned (section 13.2). Spikes, bumpers, moving platforms, lava and so on.

Rule: never pick two twists from the same category in one round (for example Moon Gravity and Heavy Boots). The v2 demo already does this.

E. Respawn and lives

Setting	Type	Default	Range	Notes
`respawn.mode`	choice	lives	off, lives, unlimited	Off = classic elimination.
`respawn.lives`	number	3	1 to 9	Used when mode is lives.
`respawn.delaySec`	number	2	1 to 5	Time before coming back.
`respawn.shieldSec`	number	1.5	0 to 3	Protection after respawn so you are not killed instantly.
`respawn.where`	choice	random_platform	random_platform, center, top	Where the player re-enters.

With respawn on, a round ends by timer (or when only one player has lives left), and the winner is decided by lives left, then by points.

F. Power-ups (see section 6)

Setting	Type	Default	Notes
`items.enabled`	on/off	on	Master switch.
`items.allowed`	multi-select	all 4	Mega Shove, Freeze, Swap, Shield.
`items.spots`	number	3	Number of power spots per round (0 to 5).
`items.firstTouch`	on/off	on	The first player to touch a spot gets the crate.
`items.keepOnKO`	on/off	off	If on, an item stays with the player after a knockout.

G. Theme (see section 7)

Setting	Type	Default	Options
`theme.mode`	choice	rotate	fixed, rotate, random, custom
`theme.fixed`	theme id	sunny	any of the 7
`theme.perRound`	list	empty	Used by custom. One theme per round.

H. Room and safety (second tab)

Setting	Type	Default	Options
`room.name`	text	"{name}'s room"	Up to 24 characters, filtered.
`room.visibility`	choice	private	private (code only), public (listed)
`room.password`	text	empty	Optional, up to 12 characters.
`room.chat`	choice	emotes	off, emotes, filtered
`room.spectators`	on/off	on	
`room.allowKick`	on/off	on	Host can kick. Vote kick later.

4.3 Rules the server must enforce

1. Only the host can change settings. Everyone else gets a read-only copy.
2. The server **validates and clamps every value** (section 13.3 has a ready function). Never trust the browser.
3. Settings are locked during a round. Some (theme, twist picker) can change between rounds.
4. Every change is broadcast as `settings:update` so lobby and host screens stay in sync.
5. The settings object has a `version` number so old clients can be handled after updates.
6. Show "Reset to preset" and "Save as my preset" (saved in the browser).

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5. Randomized arenas (a new layout every round)

Problem you found: in v2 the starting arrangement (island, platforms, where each player stands) is the same every round, so rounds feel repetitive.

Fix: generate a new arena for every round from a **seed**. Randomize five things:

What changes	How
Island position and size	Centered with +/- 50 px jitter. Size from the host setting.
Floating platforms	Count from the host setting. Positions and widths are random but always reachable.
Spawn spots	About 60 percent on the island, the rest on platforms, then **shuffled** so a player does not start in the same place twice.
Power spots	Weighted toward higher platforms (section 6).
Event timing	Which tile crumbles first, who gets the first bomb (random, seeded).

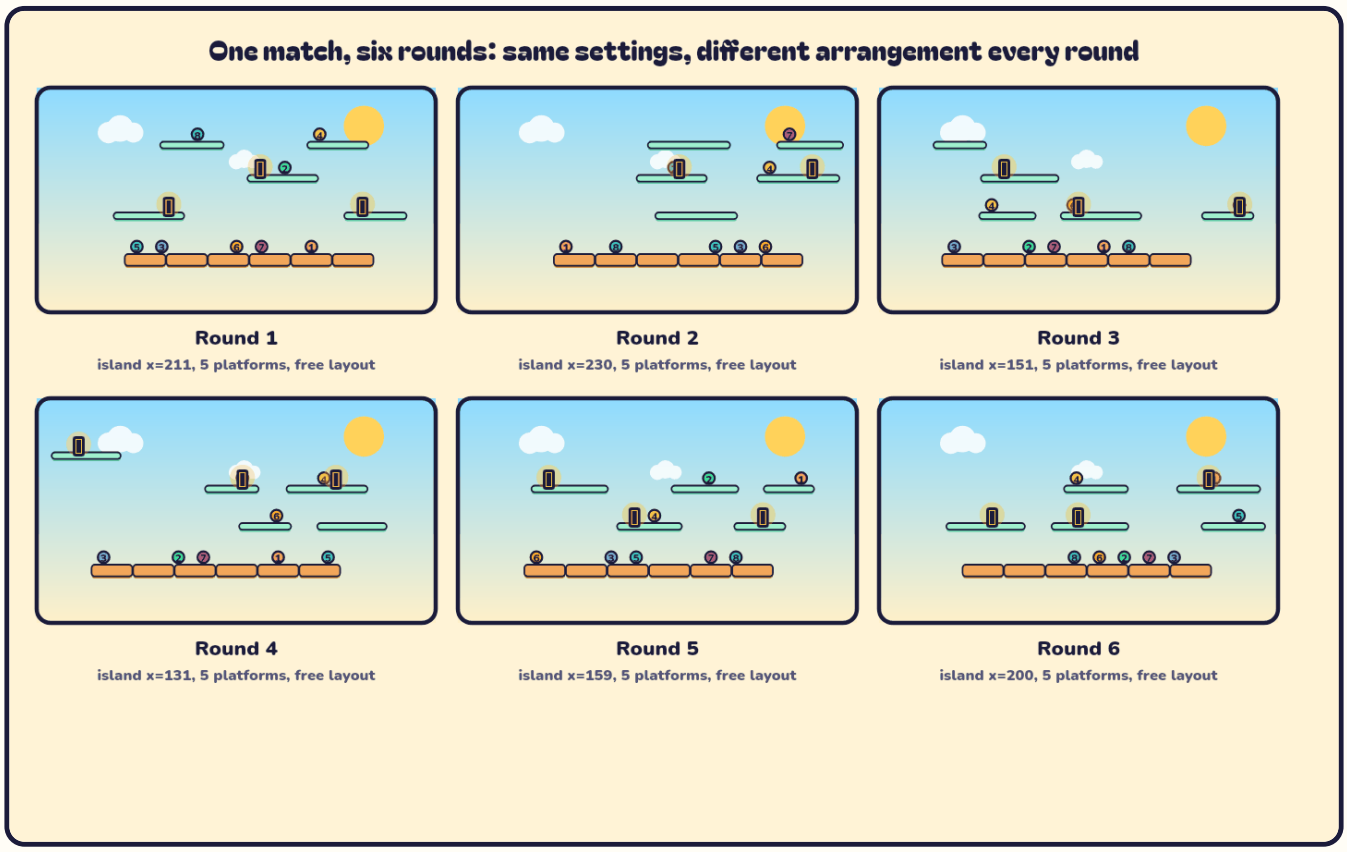


Figure 5.1: Same match, six different rounds. Numbered dots are spawn spots, stars are power spots. These were produced by the reference code in section 13.1.

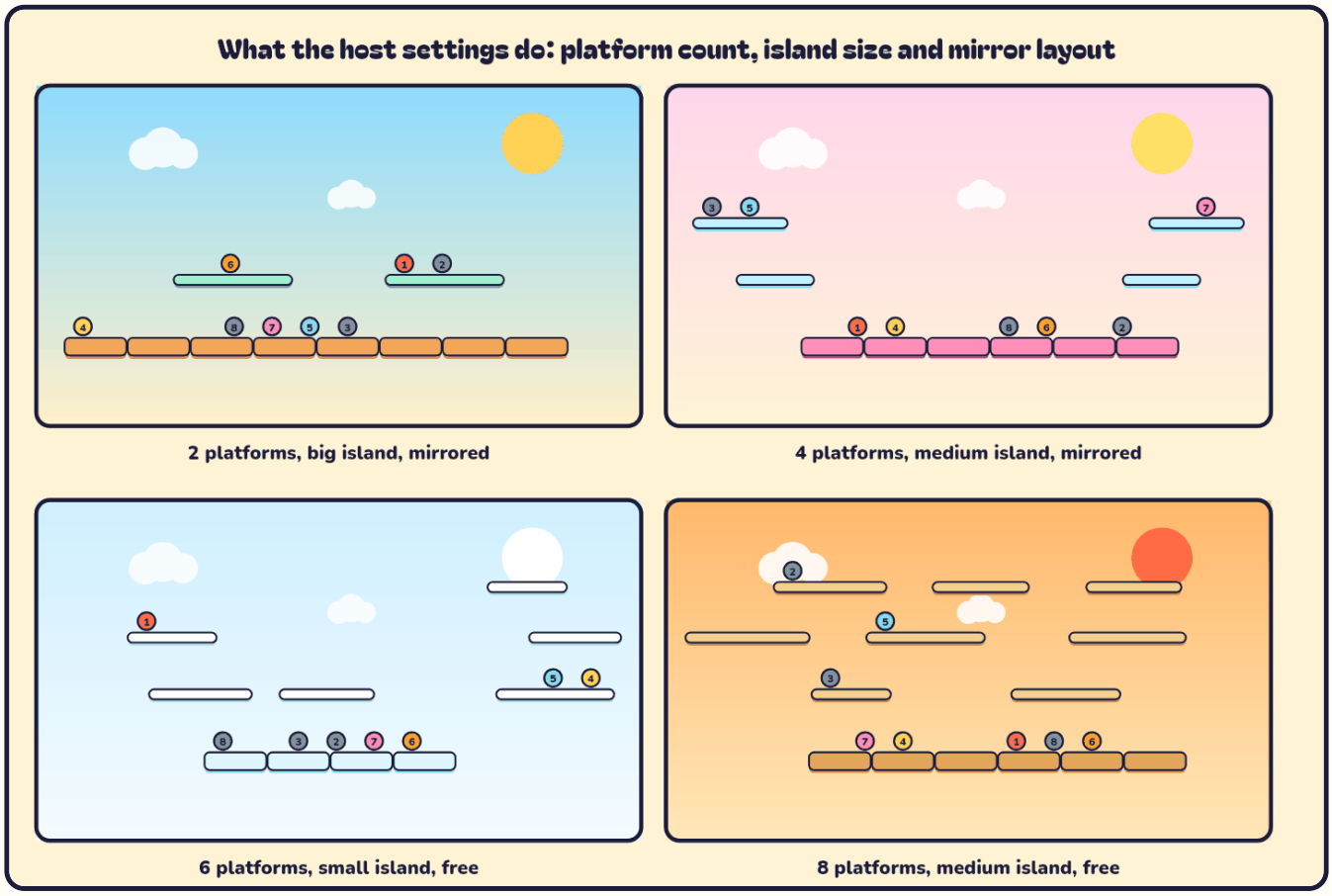


Figure 5.2: Host settings change the arena: 2, 4, 6 and 8 platforms, small and big island, mirrored or free layout, different themes.

5.1 How the generator works

1. **Seed.** The server creates a `roomSeed` when the room is created. The round seed is `hash(roomSeed, roundNumber)`. Every device gets the same numbers and builds the identical arena, so the server only has to send the seed.
2. **Separate random streams.** Layout, spawns and power spots use different streams (`layout`, `spawn`, `power`), so changing one rule does not reshuffle everything else.
3. **Island.** Width = tiles x 100 px, centered, jitter +/- 50 px.
4. **Platforms** are placed on three height tiers above the floor (100, 190 and 270 px above it). For each platform try up to 60 random positions and accept the first that passes: - It does not overlap another platform on the same tier (minimum gap 40 px). - It is not stacked within 70 px under or over another platform that overlaps it horizontally. - It is **reachable**: some existing platform is within a horizontal gap of 220 px and within a climb of 130 px.
5. **Mirror mode.** Generate half the platforms on the left half and mirror them. If the count is odd, add one centered platform.
6. **Connectivity check.** Run a breadth-first search from the island over the jump graph. If any platform cannot be reached, throw the layout away and retry with the next attempt number (up to 40 times, then fall back to the classic layout).
7. **Spawns and power spots** are chosen last (section 6.2).

5.2 Where the constants come from

They are derived from the game physics (see 13.4):

Number	Meaning	Derivation
144 px	Height of a normal jump	720 squared / (2 x 1800)
111 px	Extra height of the double jump	(720 x 0.88) squared / (2 x 1800)
130 px	Max climb used by the generator	A single jump (144) with a safety margin
336 px	Horizontal distance in one jump at full speed	420 px/s x 0.8 s air time
220 px	Max horizontal gap used by the generator	The 336 above with a safety margin

If you change gravity, jump power or speed, **recompute these constants**, or arenas may become unreachable.

5.3 Test result

The reference generator (section 13.1) was run for 8,400 combinations (platform count 2 to 8, mirrored and free, three island sizes, many seeds and rounds):

- Every arena was valid and fully reachable. Average attempts: 1.01. Fallbacks used: 0.
- The same seed, round and settings always gave the same arena.
- A different round number always gave a different arena.

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6. Power-ups and the "first to reach" rule

Your requirement: "the user will get the power when they first reach there." This is the **first-touch** rule: whoever reaches a power spot first gets the crate.

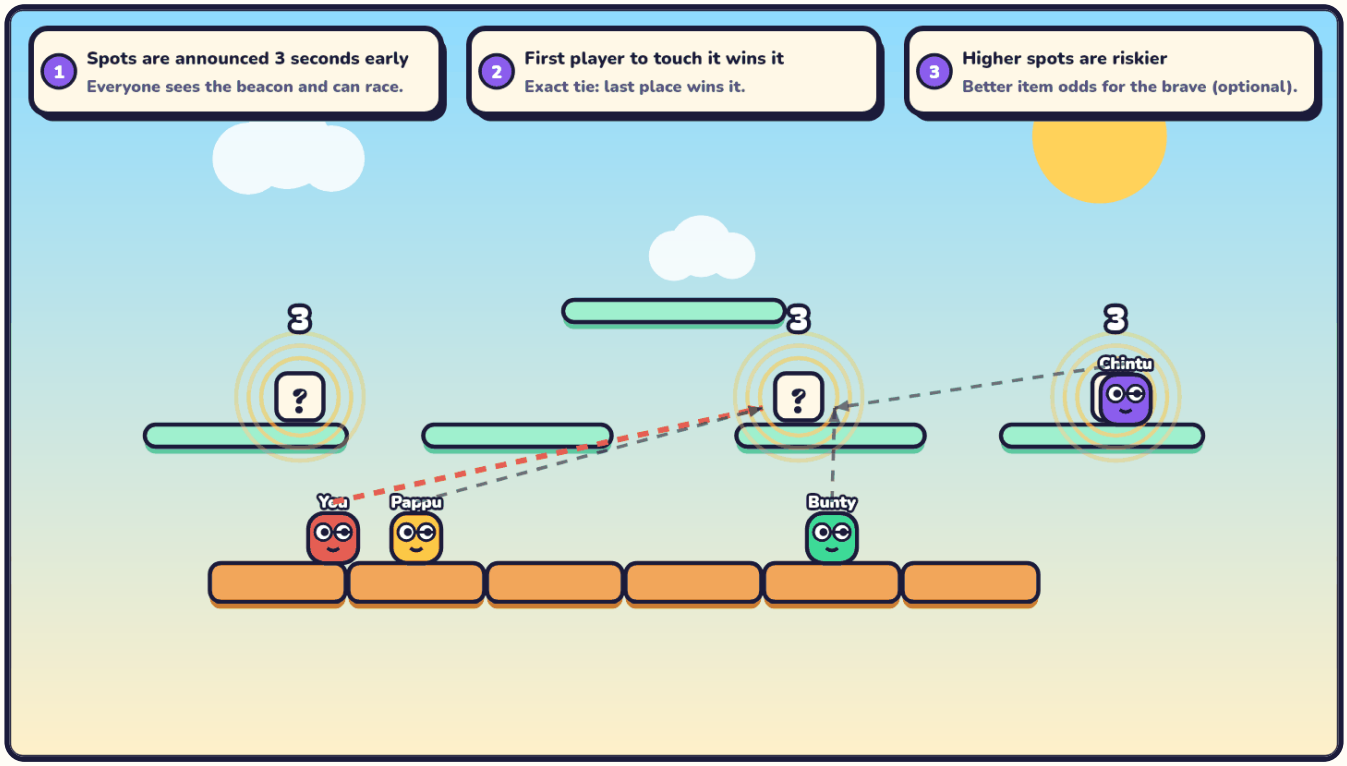


Figure 6.1: Power spots are announced with a beacon and a 3 second countdown. Everyone races. The first player to touch gets the crate.

6.1 Rules

Rule	Detail
Spots	`items.spots` fixed spots per round (default 3). Placed on platforms, **weighted toward higher ones** so the best spots are the riskiest.
Announce	A beacon with a countdown "3, 2, 1" appears at the spot before the crate drops, so everyone can race.
Pickup	First player whose body touches the crate gets it.
Tie	If two players arrive in the same server tick, the one with the **lower total score** wins (comeback help). If still tied, a seeded random choice.
Capacity	One item per player. If you hold one, touching another crate does nothing (setting: swap them).
Use	Press Shove. The item is consumed. Shield is applied instantly on pickup.
Expiry	Crates vanish after 14 seconds if nobody takes them. Items do not carry over between rounds (unless `keepOnKO`).
Refill	A spot refills 6 to 9 seconds after it was taken.

6.2 Fairness details

- **Server decides.** The server resolves pickups in tick order. The browser only shows a prediction.
- **Lag compensation.** Forgive up to 100 ms so a player on a slower network is not always beaten.
- **No spawn camping.** Spots are never placed within 200 px of each other, and never next to a spawn point of the same player twice in a row.
- **Optional "Summit Bonus".** The first player to land on the highest platform of the round gets a free item. Off by default.

6.3 Items

Item	Icon (demo)	Effect
Mega Shove	💥	Next shove has 2.6 times the force and 1.5 times the range.
Freeze Burst	🧊	Next shove freezes everyone within 170 px for 1.4 s (they cannot move).
Swap	🔀	Next shove swaps places with the nearest rival within 460 px.
Shield	🛡️	5 seconds with no knockback, no freeze, no bomb passing.

Ideas for later (not built): Magnet (pulls the crown), Banana (a slippery trap), Rocket Jump, Giant (grow for 6 s), Decoy, Reverse Curse (reverses one rival's controls).

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7. Themes and round-wise theme change

A **theme** changes colors, background, particles and music. It does **not** change gameplay unless the host turns on `twists.themeLinked`.

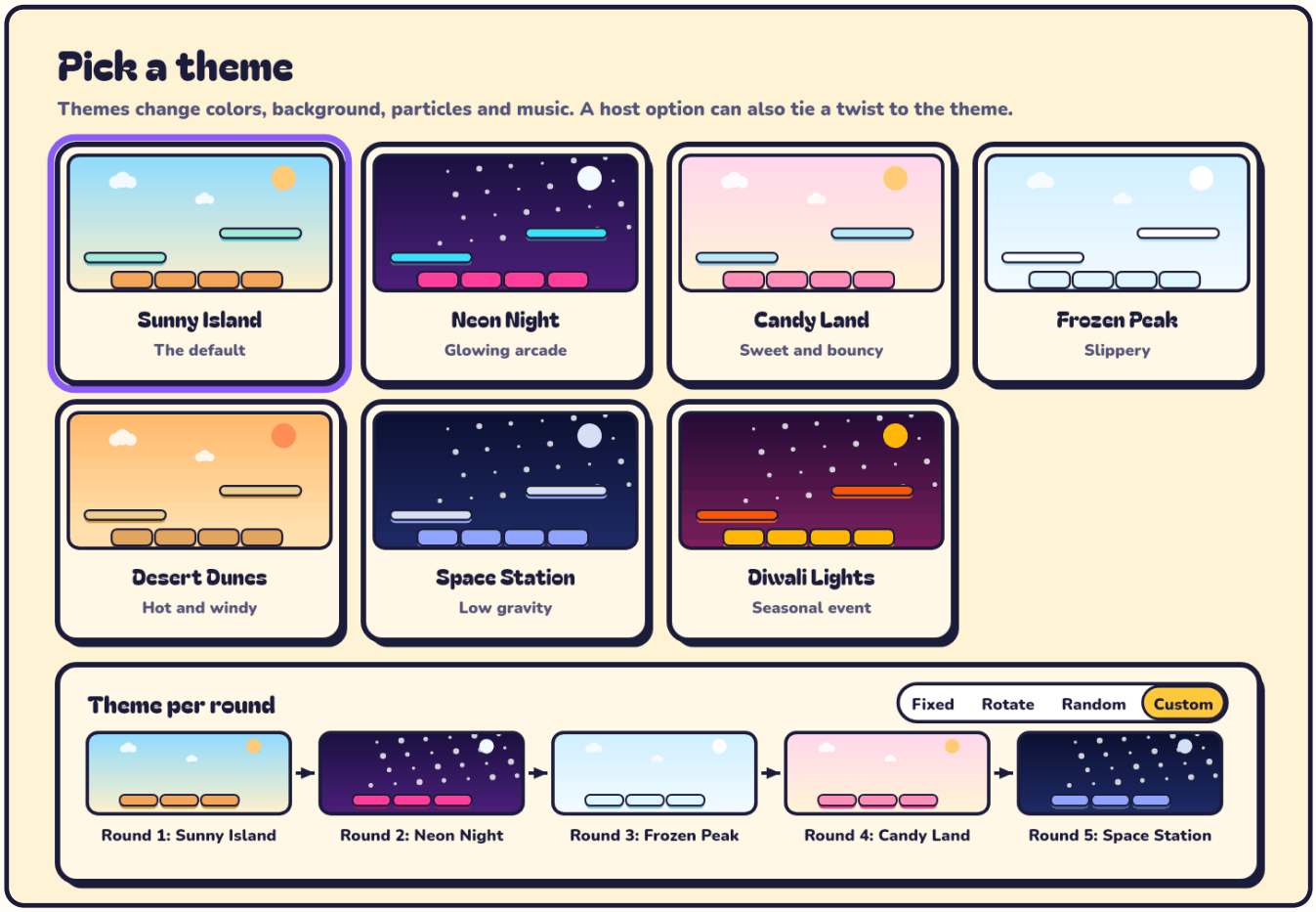


Figure 7.1 (MOCKUP): Seven themes. Below, a "Theme per round" strip lets the host set a different theme for each round, or use Fixed, Rotate or Random.

7.1 Theme modes

Mode	Behaviour
Fixed	One theme for the whole match.
Rotate	Go through the themes in order, one per round.
Random	A random theme every round, no repeat twice in a row.
Custom	The host picks a theme for each round on the strip.

7.2 The seven themes (all colors are hex values)

Theme id	Name	Sky top	Sky bottom	Sun / Moon	Floor fill / edge	Platform fill / edge	Linked twist (if enabled)
`sunny`	Sunny Island	`#8EDBFF`	`#FFF0C9`	`#FFD25A`	`#F2A65A` / `#C97A2C`	`#9FF0CE` / `#5DC9A0`	none
`neon`	Neon Night	`#1A1040`	`#4B1F7A`	`#F5F0FF`	`#FF3D9A` / `#B0206B`	`#22E3FF` / `#0B9BB5`	Lights Out
`candy`	Candy Land	`#FFD6EC`	`#FFF4D6`	`#FFE066`	`#FF8FB8` / `#D9578A`	`#B8F0FF` / `#7CCBE0`	Trampoline

Theme id	Name	Sky top	Sky bottom	Sun / Moon	Floor fill / edge	Platform fill / edge	Linked twist (if enabled)
`frozen`	Frozen Peak	`#CFEFFF`	`#F4FBFF`	`#FFFFFF`	`#DFF6FF` / `#8FD0EA`	`#FFFFFF` / `#B7DDEE`	Ice Rink
`desert`	Desert Dunes	`#FFB86B`	`#FFE7B8`	`#FF6B3D`	`#E2A55A` / `#B97A2E`	`#F4D08C` / `#C9A04F`	Gale Force
`space`	Space Station	`#0B1030`	`#1F2A66`	`#D7DEFF`	`#8FA3FF` / `#4D5FD1`	`#D7DEFF` / `#8A97E6`	Moon Gravity
`diwali`	Diwali Lights	`#240B36`	`#7A1F5C`	`#FFB703`	`#FFB703` / `#C47F00`	`#FB5607` / `#B13A00`	none (seasonal)

Only `sunny` exists in the demo. The other six are designed here and drawn in the mockup; the colors in the table are the ones used in Figure 7.1.

7.3 What a theme contains (data shape)

```
{
  "id": "neon",
  "name": "Neon Night",
  "sky": ["#1A1040", "#4B1F7A"],
  "orb": "#F5F0FF",
  "stars": true,
  "floor": ["#FF3D9A", "#B0206B"],
  "platform": ["#22E3FF", "#0B9BB5"],
  "ink": "#1B1B3A",
  "particles": "sparkle",
  "music": "neon-loop-01",
  "linkedTwist": "dark"
}
```

7.4 Rules

- Keep **ink** (the dark outline) the same in every theme so the game always feels like the same game.
- Check contrast: player colors must stay readable on every sky. Test each theme with all 8 player colors.
- Seasonal themes (Diwali, Holi, an IPL-style cricket theme, college fest) are a content and revenue opportunity (section 10). Sponsored themes must be clearly marked.
- Night themes should avoid fast flashing. Cap flashes at 3 per second (accessibility, section 11).

---

8. Design system: fonts, colors, icons, sound, motion

Style name: "Sticker Pop". Thick dark outlines, hard offset shadows, rounded squircle shapes, candy-bright colors on cream paper panels. It looks like stickers on a sheet.

8.1 Fonts

Role	Font	Weights used	Source and license	Where used
Display (titles, timer, buttons)	**Bagel Fat One**	400 (only weight)	Google Fonts, SIL Open Font License 1.1	Game title, round title, timer, card names, Play button

Role	Font	Weights used	Source and license	Where used
Body and UI	**Nunito**	700 and 900	Google Fonts, SIL Open Font License 1.1	Labels, scoreboard, chips, hints, names, feed

CSS stacks used in the demo:

```
--font-display: "Bagel Fat One", "Arial Rounded MT Bold", system-ui, sans-serif;
--font-body:    Nunito, system-ui, -apple-system, "Segoe UI", sans-serif;
```

Type scale (in game pixels on the 960 x 540 canvas): title 58, timer 46, round title 40, banner 60, button 26, card name 21, body 16, chip 15, small 13, label 12.

Production advice: download the fonts and **host them yourself** as `.woff2` instead of loading from Google. It is faster, avoids a layout jump, and avoids sending visitors' IP addresses to a third party (a privacy point under European rules). Hinglish written in English letters needs only Latin glyphs. If you add Devanagari text later, add "Noto Sans Devanagari" as a fallback.

8.2 Color tokens (the "Sticker Pop" palette)

Core tokens

Token	Name	Hex	Used for
`--mp-ink`	Ink Navy	`#1B1B3A`	All outlines, text on light panels, hard shadows
`--mp-paper`	Paper Cream	`#FFF8E7`	Panels and cards
`--mp-butter`	Butter	`#FFF3D6`	Page background (light mode)
`--mp-night`	Night Indigo	`#17172E`	Page background (dark mode)
`--mp-muted`	Slate Muted	`#55557A`	Secondary text on cream
`--mp-muted-page`	Page Muted	`#6A6A8A`	Hint text under the canvas
`--mp-sun`	Sun Yellow	`#FFC93C`	Primary buttons, selected state, "YOU" marker
`--mp-mint`	Mint	`#3DDC97`	On / success / ready
`--mp-coral`	Coral	`#FF6B6B`	Danger, out, red states
`--mp-grape`	Grape	`#8B5CF6`	Badges, focus ring
`--mp-aqua`	Aqua	`#4CC9F0`	Player color, ice accents
`--mp-danger`	Danger Red	`#FF4D4D`	High damage text, bomb ring
`--mp-ok`	Success Green	`#2A9D6A`	Points gained text
`--mp-shield`	Shield Blue	`#7DD3FC`	Shield bubble
`--mp-gold-glow`	Crown Glow	`rgba(255,201,60,.35)`	Crown glow

Player colors (index 0 is always the local player in the demo)

#	Name	Hex
0	Coral	`#FF6B6B`
1	Mint	`#3DDC97`

#	Name	Hex
2	Grape	`#8B5CF6`
3	Sun Yellow	`#FFC93C`
4	Aqua	`#4CC9F0`
5	Orange	`#FF9F1C`
6	Pink	`#F15BB5`
7	Teal	`#2EC4B6`

World colors (Sunny Island theme)

Element	Fill	Edge
Island floor tile	`#F2A65A`	`#C97A2C`
Floating platform	`#9FF0CE`	`#5DC9A0`
Ice floor (Ice Rink twist)	`#C8F0FF`	`#8FD0EA`
Ice platform	`#D7F5FF`	`#8FD0EA`
Trampoline floor	`#FF9DBB`	`#E0668C`
About-to-fall tile (blinks)	`#FF6B6B`	n/a
Sky gradient	`#8EDBFF` to `#FFF0C9`	n/a
Danger glow at screen bottom	`rgba(255,107,107,0)` to `rgba(255,107,107,.45)`	n/a

Damage color ramp: under 30 percent white, 30 to 60 percent orange `#FF9F1C`, above 60 percent red `#FF4D4D`. (Color alone is not enough for color-blind players, so the number is always shown too.)

Ready-to-paste CSS variables

```
:root {
  --mp-ink:#1B1B3A; --mp-paper:#FFF8E7; --mp-butter:#FFF3D6; --mp-night:#17172E;
  --mp-muted:#55557A; --mp-sun:#FFC93C; --mp-mint:#3DDC97; --mp-coral:#FF6B6B;
  --mp-grape:#8B5CF6; --mp-aqua:#4CC9F0; --mp-danger:#FF4D4D; --mp-ok:#2A9D6A;
  --mp-shield:#7DD3FC;
  --mp-outline:3px; --mp-radius-panel:18px; --mp-radius-chip:14px; --mp-shadow:6px 7px 0 var(--mp-ink);
}
```

8.3 Icons

What the demo uses today: Unicode emoji drawn by the phone or computer's own emoji font (Apple Color Emoji, Noto Color Emoji, Segoe UI Emoji). It needs no files, but **the same emoji looks different on each device** and a few older phones show empty boxes.

Where	Icon	Meaning
Modes	🥊 💣 👑	Last One Standing, Bomb Tag, Crown Rush
Power-ups	💥 🧊 🔀 🛡️	Mega Shove, Freeze Burst, Swap, Shield
Twists	🌙 ⚓ 🐭 🦣 🎲 🙃 🧊 ⚡ 🐦‍⬛ 🕳️ 💨 🔦 🦘	Moon Gravity, Heavy Boots, Tiny Tots, Big Boys, Size Lottery, Reversed Controls, Ice Rink, Turbo, Flappy Mode, Crumbling Floor, Gale Force, Lights Out, Trampoline

Where	Icon	Meaning
Lobby	👑🤖🎲	Host, Bot, Random name

Recommended production icon packages (check the license text before shipping):

Use	Package	License (as far as we know)	Why
Game icons that look the same on every device	**Noto Emoji** (Google) or **Twemoji** as SVG/PNG sprites	Noto Emoji: Apache 2.0 / OFL. Twemoji graphics: CC BY 4.0 (needs attribution)	Same look everywhere.
UI controls (settings, copy, share, close, kick, volume, check)	**Lucide** icons (`lucide` or `lucide-static` on npm)	ISC	Clean, consistent outline icons, tiny SVGs.
Flutter version (if you ever port)	`lucide_icons` package	ISC (verify)	Same set.

If Lucide has a matching icon for a game concept, prefer it for menus (for example bomb, crown, shield, snowflake, shuffle, moon, wind, anchor, zap). Confirm exact icon names on the Lucide website before coding.

Game art: everything inside the arena (platforms, players, crates, rings) is drawn with code on the canvas. There are **no image files** in the demo, which keeps it tiny (about 50 KB) and fast.

8.4 Sound

- **Today:** synthesized with the browser's Web Audio API (square, triangle, sawtooth beeps). No audio files.
- **Rules:** sound starts only after the first tap or key press (browsers require it). Mute on `M` and a visible mute button. Remember the mute choice.
- **Production:** add a small sound pack. Kenney.nl audio packs are CC0 (free for any use, but verify each pack). Add one looping music track per theme, quiet by default.

8.5 Shape, spacing and motion tokens

Token	Value
Outline	3 to 4 px Ink Navy on every shape
Hard shadow	offset (6 px, 7 px), solid Ink Navy, no blur
Panel radius	18 px
Chip radius	14 px (pill)
Button	full pill (radius = half height), 4 px outline
Player body	squircle, corner radius 32 percent of size, base size 36 px (hs 18)
Canvas	logical 960 x 540 (16:9), device pixel ratio capped at 2
Page border	3 px Ink Navy, radius 14 px
Pulse	scale 1 plus or minus 0.04 at about 5 Hz (Play button)
Title wave	each letter bobs 6 px, off when "reduce motion" is on
Hit-stop	70 ms freeze on a hit (120 ms for Mega Shove)
Slow motion	25 percent speed on the final knockout, back to normal in about half a second

Token	Value
Screen shake	5 to 18 px, decays quickly

8.6 HUD map

Top-left: round number and mode chip. Top-center: timer and twist chips. Top-right: scoreboard (round totals, or crown seconds in Crown Rush). Left, below the mode chip: the event feed (last 4 lines, fade after 2.6 s). Over each player: name and damage percent, plus an icon for bomb, crown or held item. The local player has a yellow arrow.

9. Technical architecture

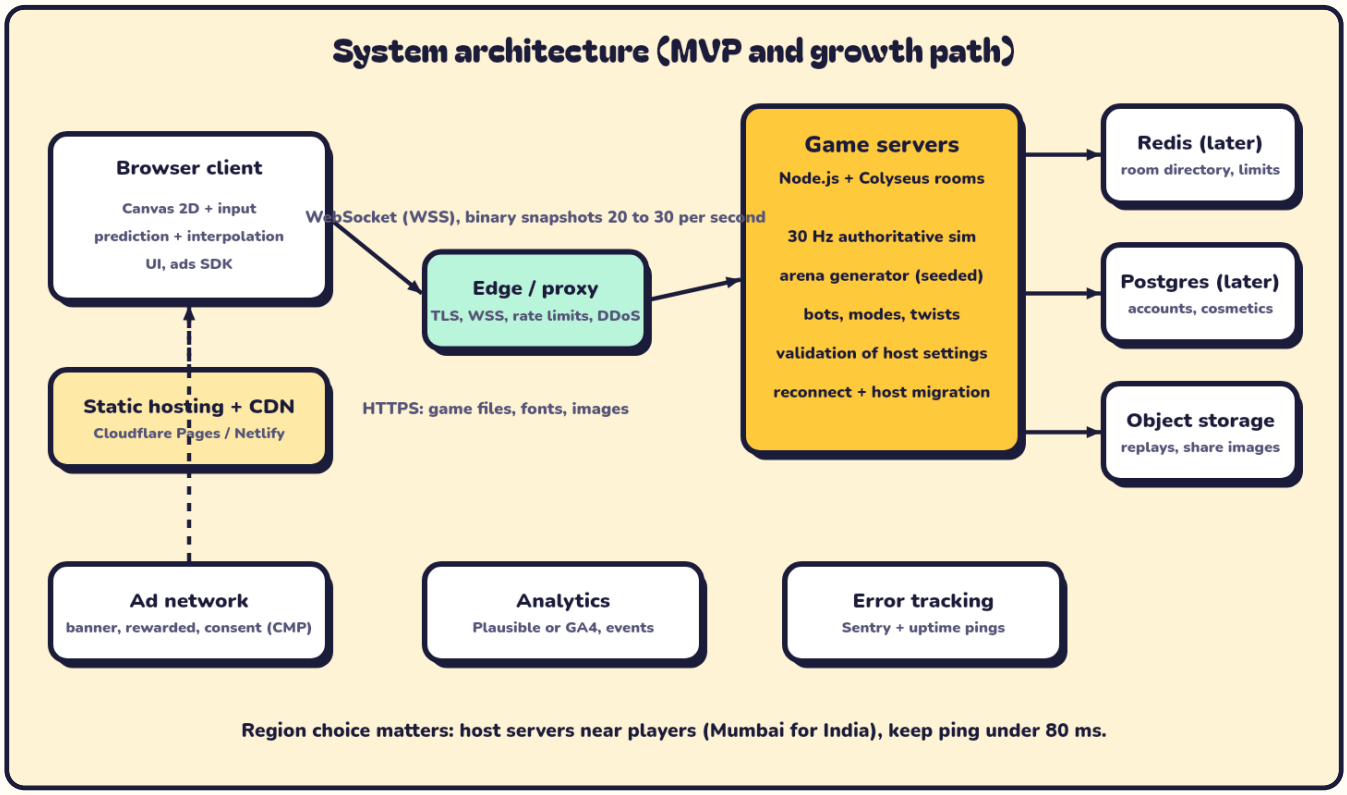


Figure 9.1: Architecture for the MVP, with later additions marked "later".

9.1 Recommended stack

Layer	Choice	Reason
Client	Keep vanilla **Canvas 2D** (what the demo uses). Optional: Phaser 3 if you want sprite tools.	The demo already works. No framework needed.
Server	**Node.js 22 + Colyseus** (rooms, state sync, reconnection built in)	Fast to build in JavaScript, the same language as the client. Alternative: plain `ws` plus your own room code.
Static files	Cloudflare Pages, Netlify or Vercel	Free tiers, global CDN.
Game servers	Fly.io, Railway or a small VPS	Pick a region near players (Mumbai for India). Verify current region list and prices.

Layer	Choice	Reason
Later	Redis for room directory and rate limits. Postgres (for example Supabase) for accounts and cosmetics.	Not needed for the MVP.
Monitoring	Sentry (errors), Plausible or GA4 (events), an uptime ping	You must see problems before players complain.

9.2 How the game stays fair and in sync

Topic	Decision
Who runs the physics?	The **server**. It is the only source of truth (this stops cheating).
Tick rate	Simulation at 30 Hz. Send snapshots 20 to 30 times a second.
Local player	Client-side prediction: move instantly on your screen, correct if the server disagrees.
Other players	Draw them about 100 ms in the past and interpolate between two snapshots, so movement is smooth.
Arena	Send only the **seed**. Every client builds the same arena (section 5).
Cosmetic effects	Screen shake, hit-stop and slow motion are client-only visuals. Do not slow the server.
Round state	A server state machine: lobby, intro, pick, play, results, final.
Bots	Run on the server using the same input format as humans.
Random numbers	Server-side `Math.random` only for gameplay events. Seeded generators for arenas.

9.3 Messages (summary)

Direction	Message	Content
Client to server	`join`	roomCode, name, color, playerId, sessionToken
Client to server	`input`	left, right, jump, shove as a small bitfield plus sequence number
Client to server	`settings:update`	host only, whole or partial settings object
Client to server	`ready`, `start`, `pick`, `vote`, `emote`, `kick`	small actions
Server to client	`state`	snapshot: positions, velocities, timers, items, bomb and crown
Server to client	`event`	KO, bomb pass, crown steal, pickup, toast text
Server to client	`round:start`	mode, twists, theme, seed, spawn slot for you
Server to client	`settings`	the validated settings

Direction	Message	Content
Server to client	`error`	human-readable reason (room full, wrong code, banned)

9.4 Rough capacity math

One snapshot is 8 players x about 16 bytes = about 130 bytes, 30 times a second = about 4 KB/s per player. A full room of 8 receives about 4 KB/s each, so about 31 KB/s per room going out. That is tiny. The limit is more likely CPU for physics and bots than bandwidth, so **measure** with a load test before promising numbers (section 13.8).

9.5 Security basics

- Server decides everything. Validate and clamp every input (speed, rate of actions).
- Room codes: 5 characters from 31 symbols is about 28.6 million codes. Rate-limit join attempts per IP and expire empty rooms quickly.
- Passwords stored only in memory for the life of the room.
- HTTPS and WSS only. Check the `Origin` header on WebSocket connections.
- Cap rooms per IP to stop room spam.

10. How to make money

Golden rule: never put ads over gameplay. A player who loses a fight because of a pop-up leaves forever. Show ads only in menus, lobby and between matches. Protect trust first, earn second.

10.1 Where ads go

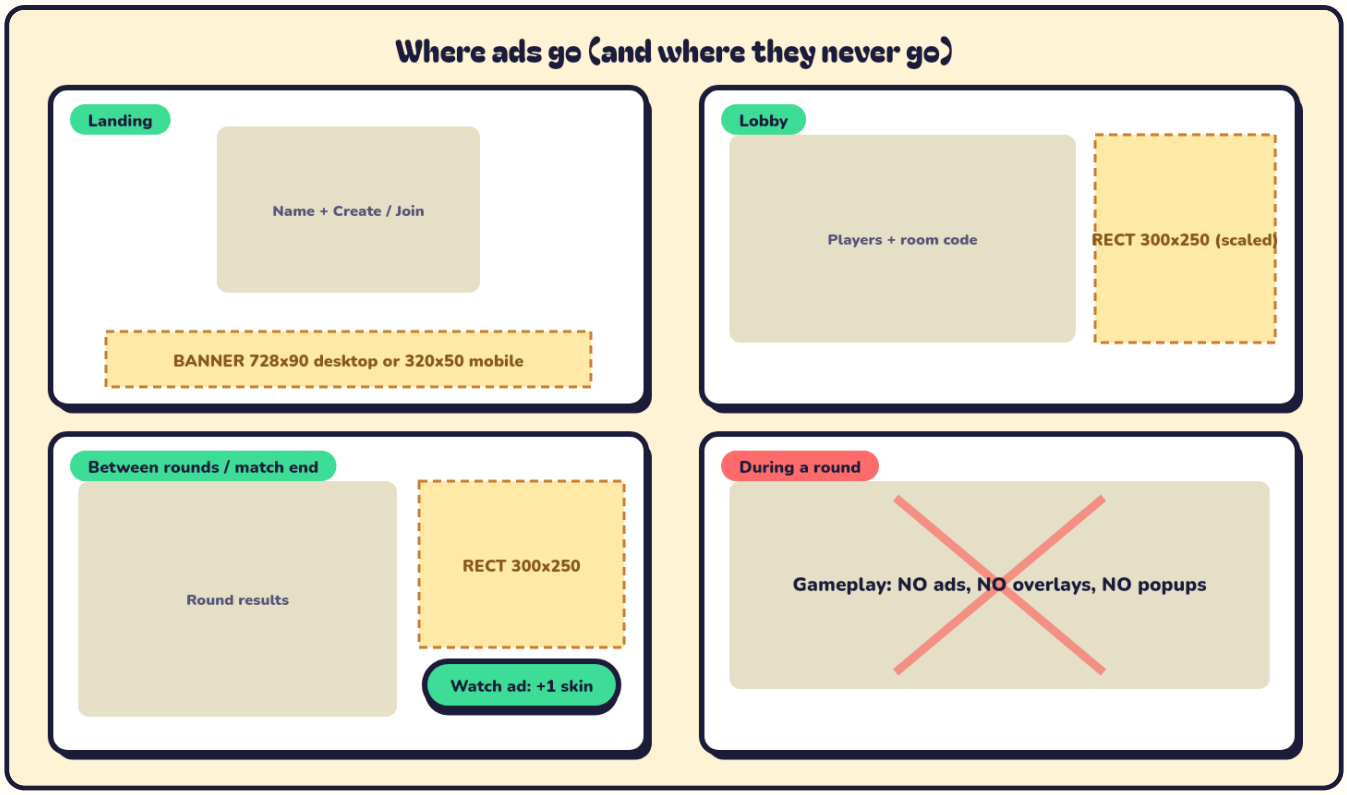


Figure 10.1: Ad slots on landing, lobby and results. Never during a round.

Screen	Ad type	Size	Notes
Landing	Banner	728x90 desktop, 320x50 mobile	Below the name box. Never next to the Play or Create button (accidental clicks are against ad rules).
Lobby	Rectangle	300x250 (scaled)	Side column while people wait.
Between rounds	Rectangle	300x250	Only on the results panel.
Match end	Interstitial	full screen	Maximum one per 8 to 10 minutes. Never in the first session. Always after a full match.
Anywhere	Rewarded video	opt-in	"Watch an ad to unlock a skin for 1 day." The player chooses.
During a round	**None**		No ads, no overlays, no pop-ups.

10.2 Ad networks and platforms (verify current terms)

Option	What it is	Notes
Google AdSense / Ad Manager	Banner and display ads on your website	Needs real content pages (how to play, news, FAQ) to be approved. A single game screen can be rejected as thin content. Follow the placement policies.
Google H5 Games Ads (Ad Placement API)	Interstitial and rewarded ads made for browser games	Needs approval. Check eligibility.

Option	What it is	Notes
Game portals (CrazyGames, Poki, GameDistribution and similar)	They host your game and share ad revenue	You get their audience but give up part of the revenue and need to follow their rules, including for online multiplayer. Check each portal's terms.
Mobile app route (AdMob)	If you wrap the game as an Android app	More work, higher reach on phones.
Ad mediation (for example AppLovin or similar)	Several networks compete for each impression	Raises earnings once you have steady traffic.

10.3 Other income

Idea	How	Notes
Cosmetics	Skins, hats, trails, victory poses, emote packs	The best long-term income. Never sell power.
Season pass	A monthly reward track	Needs accounts. Later.
Host Pro	Subscription: bigger rooms (16+), custom themes, no ads, private tournaments	Good for teachers, streamers, colleges.
Sponsored themes and crates	A brand skins a theme or a crate for a campaign, for example a cricket league, a snack brand, a college fest	Label clearly as sponsored.
College and fest rooms	Branded leaderboard, event codes	Fits a student audience.
Tips	UPI QR code, "Buy me a coffee"	Small but easy.
Merch	Stickers (the style suits it)	Later.
Creator code	Streamers get a code to share, you share revenue	Growth plus income.

India notes: UPI is the payment people already use. Payment gateways such as Razorpay or Cashfree support it. Talk to a chartered accountant about GST and income tax on digital income before you take money.

10.4 What revenue can look like (illustrative only)

The numbers below are **assumptions, not predictions**. Real eCPM (earnings per 1,000 views) varies a lot by country, time of year, device and ad network. Replace them with numbers from your own ad dashboard after a few weeks.

Formula

```
Banner and rectangle revenue / month = DAU x sessions per day x ad views per session x eCPM / 1000 x 30
Rewarded revenue / month             = DAU x rewarded views per user per day x rewarded eCPM / 1000 x 30
Cosmetics / month                    = DAU x 4 (to estimate monthly users) x paying share x average spend
```

Assumptions used: 2.5 sessions per user per day, 6 ad views per session, 0.3 rewarded views per user per day, monthly users = 4 x daily users, 0.5 percent of monthly users pay, average spend 2 USD per month.

Scenario	Daily users	Banner eCPM	Rewarded eCPM	Banner / month	Rewarded / month	Cosmetics / month	Total / month
Low	1,000	0.30 USD	4 USD	about 135 USD	about 36 USD	about 40 USD	about 211 USD
Base	10,000	0.80 USD	8 USD	about 3,600 USD	about 720 USD	about 400 USD	about 4,720 USD
High	50,000	2.00 USD	15 USD	about 45,000 USD	about 6,750 USD	about 2,000 USD	about 53,750 USD

Reality check: reaching 10,000 daily users is hard and takes months of growth work. Hosting is small at first (roughly tens of dollars a month at the low end, more as rooms grow; measure). The honest early goal is **retention and word of mouth**, not revenue. Add ads only once players come back on day 7.

10.5 What not to do

- No ads inside gameplay, no fake "close" buttons, no ads that look like game buttons.
- No selling power-ups that win matches (pay-to-win kills rooms with friends).
- No collecting personal data you do not need.
- No ads shown to players you believe are children without checking the network's child-directed rules (section 11.1).

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11. Things you may have forgotten (care checklist)

Labels: **MUST** = before any public launch. **SHOULD** = before real growth. **LATER** = after traction.

11.1 Legal, privacy and safety

Level	Item	Notes
MUST	Privacy Policy and Terms of Use pages	Required by ad networks and app stores. Say what you collect (name, IP, cookies, analytics).
MUST	Children and teenagers	Many players are under 18. India's Digital Personal Data Protection Act, 2023 treats under-18s as children and restricts tracking and targeted ads for them (the rules are being phased in; confirm the current status with a lawyer). The US has COPPA (under 13) and Europe has GDPR rules. Safest design: no accounts, no personal data, **non-personalized (contextual) ads** and the ad network's child-directed or "under age of consent" tags where required.
MUST	Cookie and ad consent for Europe	Use a Google-certified consent tool (CMP) if you serve ads to EU or UK visitors.
MUST	Moderation of names and room names	Server-side filter, report button, host kick.
MUST	No free-text chat at launch	Use emotes only. Free chat needs moderation, reporting and more legal work.

Level	Item	Notes
MUST	Name and trademark check	"Mutator Party" must not clash with existing games. Search the Indian trademark registry and the web. Do not use "Skribbl" or "Smash Karts" or look-alike art.
MUST	Asset licenses	Fonts are OFL (fine). Emoji packs may need attribution (Twemoji CC BY). Keep a `CREDITS` page.
MUST	Employment and IP	If you work or are about to work for a games or tech company, read the outside-work and IP clauses in your contract **before** launching a commercial game. Get written permission if needed.
SHOULD	Grievance and contact e-mail	A visible address for complaints, takedowns and parents.
SHOULD	Age notice	A plain line such as "Meant for players 13 and over" with parental guidance wording reviewed by a lawyer.
SHOULD	Company or registration	Decide how income will be received and taxed (talk to a chartered accountant).
LATER	Accounts	Add Google or phone sign-in only when you sell cosmetics. Then data protection duties grow.

11.2 Technical

Level	Item	Notes
MUST	Use `event.code` for movement keys	Letters differ on non-QWERTY keyboards. `KeyA`, `KeyD`, `KeyW` are position-based.
MUST	Pause or throttle in background tabs	Browsers slow hidden tabs. The server must never depend on client timers.
MUST	Phone support	Landscape layout, safe-area padding, big touch buttons (at least 48 px), multi-touch for move plus jump, no text selection, no pull-to-refresh.
MUST	Audio unlock	Start sound after the first tap (Safari and Chrome rules).
MUST	Server validation	Clamp every host setting and every input (section 13.3).
MUST	Reconnect	30 second grace, bot takes over (section 3.3).
MUST	Performance budget	Stable 60 fps on a cheap Android phone. Cap particles (for example 300), keep draw calls low, cap device pixel ratio at 2.
SHOULD	Offline and install	Make it a PWA (installable, cached files).
SHOULD	Share cards	Open Graph image and title so links look good in WhatsApp, Discord and X.
SHOULD	Error tracking	Sentry or similar from day one.
SHOULD	Versioning	Version the settings object and the network protocol. A deploy must not crash live rooms (drain old rooms, then restart).
SHOULD	Time and clock	Show server-provided countdowns. Never trust the player's clock.
LATER	Gamepad support	Standard Gamepad API.
LATER	Replays and clips	Record the last 8 seconds of events and export a short video.

11.3 Gameplay and fairness

Level	Item	Notes
MUST	Fix round length	See 2.4. Rounds must last 40 to 75 seconds.
MUST	Tie-breaks	Define what happens for ties in points, in crown seconds and at time up.
MUST	Anti-snowball and comeback	Last place picks the twist (done). Consider slightly stronger first-touch priority for last place.
MUST	Banned twist combinations	Test all 13 twists in pairs and triples. Remove or limit unfun mixes (for example Flappy plus Moon Gravity may be too floaty, Heavy plus Crumbling may be too punishing).
MUST	Spawn fairness	Mirrored layouts or equal-distance spawns so no one is boxed in.
SHOULD	Tutorial	A 20 second first-time practice with one bot.
SHOULD	Sudden death	If the timer ends with many players alive, shrink the arena.
SHOULD	Spectators	People who are out can watch and send emotes.
SHOULD	AFK handling	Section 3.3.
LATER	Teams	2 vs 2 and 4 vs 4 modes.
LATER	Hazards	Section 13.2.

11.4 Accessibility

Level	Item	Notes
MUST	Do not rely on color alone	Players also have names and different color shapes or hats. Damage shows a number.
MUST	Reduce motion option	Turns off title wave, shake, slow motion and big particle bursts.
MUST	Flash safety	No more than 3 flashes per second. Review Lights Out and explosions.
SHOULD	Rebindable keys, one-handed touch layout	
SHOULD	Text size	Minimum 12 px on the canvas, scalable.
SHOULD	Subtitles for sound cues	A visual cue for every important sound (bomb ticking, crate spawn).
SHOULD	Screen reader labels	For all menu buttons and form fields (the canvas itself cannot be read).

11.5 Growth, analytics and operations

Level	Item	Notes
MUST	One-tap invite	Copy link and WhatsApp share. Most Indian groups already live in WhatsApp.
MUST	Events to track	`room_created`, `player_joined`, `match_started`, `match_finished`, `round_length`, `ad_viewed`, `share_clicked`, `rage_quit`.
SHOULD	Key numbers	Day 1 and day 7 return rate, average session length, rounds per session, invites per player, how full rooms get.
SHOULD	Feedback button	A form or link on the results screen.

Level	Item	Notes
SHOULD	Community	A Discord or WhatsApp community for testers.
SHOULD	Launch channels	Your own Hinglish tech channel (Tech Pulse) for devlogs and launch posts, college groups and fests, Reddit communities for browser games, short vertical clips of knockouts.
SHOULD	Landing page content	A "How to play" page, FAQ and patch notes. Helps SEO and ad approval.
SHOULD	Cost alerts	Budget alerts on hosting.
LATER	A/B tests	For ad placement and onboarding.
LATER	Localization	Hinglish and Hindi text packs.

12. Roadmap and effort

Estimates assume one developer working about 10 to 12 hours a week. They are rough and should be revisited after the first milestone.

Milestone	Goal	Main tasks	Estimate
M0 (done)	Offline demo v2	3 modes, 13 twists, items, pick screen, bots.	done
M0.5	Fix feel	Tune knockback, bots, bomb pass. Hit the round length targets. Switch keys to `event.code`.	1 week
M1	Online MVP	Name prompt, rooms and codes, lobby, server physics, 3 modes, seeded random arenas, reconnect, bots on server.	5 to 7 weeks
M2	Host controls and polish	Full settings screen, validation, presets, themes (3 first), power spots with first-touch, mobile controls, tutorial.	4 to 6 weeks
M3	Public beta	Analytics, error tracking, privacy and terms pages, moderation tools, share cards, banner ads on landing and lobby, WhatsApp share.	3 to 4 weeks
M4	Retention	Cosmetics, rewarded ads, more themes (including seasonal), hazards, more modes, streamer tools.	ongoing

Definition of done for M1: 4 friends on 4 phones can join one room by link, finish 5 rounds with no desync, and any one of them can drop and rejoin.

Biggest risks

Risk	Plan
Netcode feels laggy on mobile networks	Deploy near players, prediction plus interpolation, show ping, test on throttled networks early.
Nobody finds the game	Build the share loop first (link, WhatsApp, clips) and use your own channel for launch.
Rounds are not fun	Playtest every week with real people. Keep a list of fun and not-fun moments.
Moderation or legal trouble	Emotes only, no accounts, strict name filter, privacy-first.
Scope growth	Ship M1 before adding hazards, teams or cosmetics.

13. Appendices

13.1 Reference code: arena generator (single file, copy-paste ready)

Save as `arena.js`. It has no dependencies and works in Node and in the browser. It was tested on 8,400 setting combinations (section 5.3).

```
/* arena.js - deterministic per-round arena generator (reference implementation, no dependencies).
   Same (seed, round, settings) => same arena on every client and on the server. */
function mulberry32(a){ return function(){ a |= 0; a = a + 0x6D2B79F5 | 0; let t = Math.imul(a ^ a >>> 15, 1 |
a);
  t = t + Math.imul(t ^ t >>> 7, 61 | t) ^ t; return ((t ^ t >>> 14) >>> 0) / 4294967296; }; }
function hashSeed(...parts){ let h = 2166136261 >>> 0; for (const c of parts.join('|')){ h ^= c.charCodeAt(0); h
= Math.imul(h, 16777619); } return h >>> 0; }

const ARENA = { W:960, H:540, FLOOR_Y:400, TILE_W:100, TIERS:[300, 210, 130],
  MAX_STEP_UP:130,   // single jump apex is ~144px; keep a safety margin
  MAX_GAP:220,       // air time ~0.8s * 420px/s = ~336px; keep a safety margin
  MIN_W:120, MAX_W:200, MIN_SEP:40 };

function generateArena({ seed, round = 0, platformCount = 4, islandTiles = 6, symmetric = true, players = 8,
powerSpots = 3 }){
  for (let attempt = 0; attempt < 40; attempt++){
    const rnd = mulberry32(hashSeed(seed, round, 'layout', attempt));
    const R = (a, b) => a + rnd() * (b - a);
    const A = ARENA, isl = { w:islandTiles * A.TILE_W, y:A.FLOOR_Y };
    isl.x = Math.round(A.W / 2 - isl.w / 2 + R(-50, 50));
    const plats = [], all = () => [isl, ...plats];
    const overlapX = (a, b, pad = 0) => a.x < b.x + b.w + pad && b.x < a.x + a.w + pad;
    const gapX = (a, b) => Math.max(0, Math.max(a.x - (b.x + b.w), b.x - (a.x + a.w)));
    const ok = c => {
      for (const p of all()) if (p.y === c.y && overlapX(c, p, A.MIN_SEP)) return false;
      for (const p of all()) if (overlapX(c, p) && Math.abs(p.y - c.y) < 70) return false;
      return all().some(p => gapX(c, p) <= A.MAX_GAP && (p.y - c.y) <= A.MAX_STEP_UP);
    };
    const want = symmetric ? Math.floor(platformCount / 2) : platformCount;
    for (let i = 0; i < want; i++){
      for (let tries = 0; tries < 60; tries++){
        const r = rnd(), tier = r < .45 ? 0 : r < .85 ? 1 : 2, w = Math.round(R(A.MIN_W, A.MAX_W));
        const maxX = symmetric ? A.W / 2 - 20 - w : A.W - 30 - w;
        const c = { x:Math.round(R(30, Math.max(31, maxX))), y:A.TIERS[tier], w };
        if (ok(c)){ plats.push(c); if (symmetric) plats.push({ x:A.W - c.x - c.w, y:c.y, w:c.w }); break; }
      }
    }
    if (symmetric && platformCount % 2){ const w = 160; plats.push({ x:A.W / 2 - w / 2, y:A.TIERS[1], w }); }
    // validate: every platform reachable from the island (BFS over the jump graph)
    const seen = new Set([isl]), q = [isl];
    while (q.length){ const a = q.shift(); for (const b of all()){ if (seen.has(b)) continue;
      const up = a.y - b.y, can = (gapX(a, b) <= A.MAX_GAP && up <= A.MAX_STEP_UP) || (gapX(a, b) <= A.MAX_GAP &&
up < 0 && Math.abs(up) <= 400);
      if (can){ seen.add(b); q.push(b); } } }
    if (plats.length !== platformCount || seen.size !== all().length) continue;
    // spawns: ~60% on the island, rest on platforms, then shuffled so slot -> player changes every round
    const srnd = mulberry32(hashSeed(seed, round, 'spawn'));
    const shuffle = arr => { for (let i = arr.length - 1; i > 0; i--){ const j = Math.floor(srnd() * (i + 1));
[arr[i], arr[j]] = [arr[j], arr[i]]; } return arr; };
    const spots = [];
    for (const p of all()) for (let x = p.x + 30; x <= p.x + p.w - 30; x += 60) spots.push({ x, y:p.y - 18,
onIsland:p === isl });
    const isle = shuffle(spots.filter(s => s.onIsland)), top = shuffle(spots.filter(s => !s.onIsland));
    const nIsle = Math.min(isle.length, Math.ceil(players * .6)), spawns = shuffle([...isle.slice(0,
nIsle), ...top].slice(0, players));
    // power spots: favour higher (harder to reach) platforms, keep them apart
    const prnd = mulberry32(hashSeed(seed, round, 'power')), spotsOut = [];
    const cand = plats.length ? plats : [isl];
    for (let k = 0; k < powerSpots * 12 && spotsOut.length < powerSpots; k++){
      const p = cand[Math.floor(prnd() * cand.length)], wgt = 1 + (A.FLOOR_Y - p.y) / 100;
      if (prnd() * 3.5 > wgt) continue;
```

```
      const s = { x:Math.round(p.x + 30 + prnd() * (p.w - 60)), y:p.y - 20 };
      if (spotsOut.every(o => Math.hypot(o.x - s.x, o.y - s.y) > 200)) spotsOut.push(s);
    }
    return { seed, round, island:{ x:isl.x, y:isl.y, w:isl.w, tiles:islandTiles }, platforms:plats, spawns,
powerSpots:spotsOut, attempts:attempt + 1 };
  }
  // fallback: classic layout (never fails)
  return { seed, round, island:{ x:180, y:400, w:600, tiles:6 }, platforms:[{ x:50, y:290, w:190 }, { x:720, y:
290, w:190 }, { x:385, y:195, w:190 }].slice(0, platformCount),
    spawns:Array.from({ length:players }, (_, i) => ({ x:210 + i * 77, y:382 })), powerSpots:[{ x:480, y:175 }],
attempts:41, fallback:true };
}
if (typeof module !== 'undefined') module.exports = { generateArena, mulberry32, hashSeed, ARENA };
```

Quick use:

```
const { generateArena } = require('./arena');
const arena = generateArena({ seed: 2024, round: 2, platformCount: 5, islandTiles: 6, symmetric: true, players:
8, powerSpots: 3 });
console.log(arena.island, arena.platforms.length, arena.spawns.length, arena.powerSpots);
```

13.2 Twist and hazard catalog

Twists in the demo (13)

Id	Name	Category	Effect (exact values from the code)
`low`	Moon Gravity	gravity	Gravity x0.33, jump power x0.78
`heavy`	Heavy Boots	gravity	Gravity x1.7, jump power x1.2
`tiny`	Tiny Tots	size	Everyone scale 0.55
`giant`	Big Boys	size	Everyone scale 1.7
`lotto`	Size Lottery	size	Each player gets a random scale 0.5 to 1.8
`reverse`	Reversed Controls	controls	Left and right swapped (bots too)
`ice`	Ice Rink	controls	Acceleration 420, braking 110 (almost no grip)
`speed`	Turbo	controls	Top speed x1.55
`flap`	Flappy Mode	jumping	Unlimited air jumps (each at 88 percent power)
`crumble`	Crumbling Floor	world	Island tiles fall: first after 4 s, then every 3.2 s
`wind`	Gale Force	world	A constant drift of 170 px/s that changes direction every 2.5 to 4 s
`dark`	Lights Out	world	Screen is dark. Light circles of radius 120 around players (170 for you)
`bounce`	Trampoline Land	world	Landing faster than 260 px/s bounces you back at 80 percent speed

Rule: only one twist per category per round.

Hazards (PLANNED, none built yet)

Id	Name	Effect	Build difficulty	Safety or fairness note
`bumper`	Bumpers	Round pads that bounce players	Easy	Place away from spawns.
`spikes`	Spike strips	Removes a life on touch (or knockback)	Medium	Never under a spawn point.

Id	Name	Effect	Build difficulty	Safety or fairness note
`mover`	Moving platforms	Platforms slide left and right	Medium	Predictable loop, show the path.
`lava`	Rising lava	A kill zone slowly rises	Medium	Good sudden-death tool.
`meteor`	Meteor shower	Warned circles then impacts	Medium	1 second warning ring.
`portal`	Portals	Two linked doors	Medium	Cooldown to avoid loops.
`conveyor`	Conveyor floor	A belt pushes players	Easy	Show arrows.
`shrink`	Shrinking arena	The island loses a tile every few seconds	Easy	Already half built (crumble).

13.3 Reference code: room settings defaults and server validation (single file)

Save as `settings.js`. The server runs `sanitize()` on every host change.

```
/* settings.js - room settings: defaults + server-side validation (reference implementation, no dependencies).
   Never trust the client: the server runs sanitize() on every settings:update from the host. */
const MODES = ['brawl', 'bomb', 'crown'];
const TWISTS =
['low','heavy','tiny','giant','lotto','reverse','ice','speed','flap','crumble','wind','dark','bounce'];
const ITEMS = ['mega', 'freeze', 'swap', 'shield'];
const THEMES = ['sunny','neon','candy','frozen','desert','space','diwali'];
const DEFAULTS = {
  version: 1,
  room:   { name:'', visibility:'private', password:'', chat:'emotes', spectators:true },
  match:  { maxPlayers:8, fillBots:true, botSkill:'normal', rounds:5, roundTime:'auto', scoring:'standard' },
  arena:  { platformCount:4, islandSize:'M', layout:'every_round', mirror:true, shuffleSpawns:true },
  modes:  { pool:[...MODES], order:'shuffle_bag', fixedList:[], twistPicker:'last_place' },
  twists: { allowed:[...TWISTS], perRound:'ramp', themeLinked:false },
  respawn:{ mode:'lives', lives:3, delaySec:2, shieldSec:1.5 },
  items:  { enabled:true, allowed:[...ITEMS], spots:3, firstTouch:true, keepOnKO:false },
  theme:  { mode:'rotate', fixed:'sunny', perRound:[] },
};
const clamp = (v, a, b, d) => (Number.isFinite(+v) ? Math.min(b, Math.max(a, +v)) : d);
const oneOf = (v, list, d) => (list.includes(v) ? v : d);
const subset = (arr, list, d) => { const a = Array.isArray(arr) ? [...new Set(arr.filter(x =>
list.includes(x)))] : []; return a.length ? a : d.slice(); };
function sanitize(input = {}, base = DEFAULTS){
  const i = input, o = JSON.parse(JSON.stringify(base)), g = k => (i[k] && typeof i[k] === 'object' ? i[k] : {});
  const r = g('room'), m = g('match'), a = g('arena'), md = g('modes'), t = g('twists'), rs = g('respawn'), it =
g('items'), th = g('theme');
  o.room.name = String(r.name ?? o.room.name).replace(/[^\p{L}\p{N} _'-]/gu, '').trim().slice(0, 24);
  o.room.visibility = oneOf(r.visibility, ['private', 'public'], o.room.visibility);
  o.room.password = String(r.password ?? '').slice(0, 12);
  o.room.chat = oneOf(r.chat, ['off', 'emotes', 'filtered'], o.room.chat);
  o.room.spectators = r.spectators !== false;
  o.match.maxPlayers = clamp(m.maxPlayers, 2, 12, 8);
  o.match.fillBots = m.fillBots !== false;
  o.match.botSkill = oneOf(m.botSkill, ['easy', 'normal', 'hard'], 'normal');
  o.match.rounds = clamp(m.rounds, 1, 15, 5);
  o.match.roundTime = m.roundTime === 'auto' || m.roundTime == null ? 'auto' : clamp(m.roundTime, 30, 120,
'auto');
  o.match.scoring = oneOf(m.scoring, ['standard', 'winner_plus', 'kills'], 'standard');
  o.arena.platformCount = Math.round(clamp(a.platformCount, 2, 8, 4));
  o.arena.islandSize = oneOf(a.islandSize, ['S', 'M', 'L'], 'M');
  o.arena.layout = oneOf(a.layout, ['every_round', 'per_match', 'classic'], 'every_round');
  o.arena.mirror = a.mirror !== false;
  o.arena.shuffleSpawns = a.shuffleSpawns !== false;
  o.modes.pool = subset(md.pool, MODES, MODES);
  o.modes.order = oneOf(md.order, ['shuffle_bag', 'random', 'fixed'], 'shuffle_bag');
  o.modes.fixedList = Array.isArray(md.fixedList) ? md.fixedList.filter(x => MODES.includes(x)).slice(0, 15) :
[];
  o.modes.twistPicker = oneOf(md.twistPicker, ['last_place', 'vote', 'host', 'random', 'round_robin'],
'last_place');
  o.twists.allowed = Array.isArray(t.allowed) ? [...new Set(t.allowed.filter(x => TWISTS.includes(x)))] :
```

```
TWISTS.slice(); // empty = no twists, allowed
  o.twists.perRound = oneOf(t.perRound, ['1', '2', '3', 'ramp'], 'ramp');
  o.twists.themeLinked = t.themeLinked === true;
  o.respawn.mode = oneOf(rs.mode, ['off', 'lives', 'unlimited'], 'lives');
  o.respawn.lives = Math.round(clamp(rs.lives, 1, 9, 3));
  o.respawn.delaySec = clamp(rs.delaySec, 1, 5, 2);
  o.respawn.shieldSec = clamp(rs.shieldSec, 0, 3, 1.5);
  o.items.enabled = it.enabled !== false;
  o.items.allowed = Array.isArray(it.allowed) ? [...new Set(it.allowed.filter(x => ITEMS.includes(x)))] :
ITEMS.slice();
  o.items.spots = Math.round(clamp(it.spots, 0, 5, 3));
  o.items.firstTouch = it.firstTouch !== false;
  o.items.keepOnKO = it.keepOnKO === true;
  o.theme.mode = oneOf(th.mode, ['fixed', 'rotate', 'random', 'custom'], 'rotate');
  o.theme.fixed = oneOf(th.fixed, THEMES, 'sunny');
  o.theme.perRound = Array.isArray(th.perRound) ? th.perRound.filter(x => THEMES.includes(x)).slice(0, 15) : [];
  return o;
}
if (typeof module !== 'undefined') module.exports = { DEFAULTS, sanitize, MODES, TWISTS, ITEMS, THEMES };
```

Test run (real output): `sanitize({})` returned exactly the defaults. A hostile input (`maxPlayers: 9999`, `rounds: -5`, a script tag as round time, an unknown mode `nuke`, `platformCount: '100'`, a bad theme id, `lives: 0`) was clamped to 12 players, 1 round, `auto` time, all three real modes, 8 platforms, a valid theme list, 1 life.

13.4 Physics constants (from the demo code)

Constant	Value	Meaning
Canvas	960 x 540	Logical size
`GRAV`	1800 px/s squared	Gravity
`MOVE`	420 px/s	Top walking speed
`JUMP`	720 px/s	Jump launch speed (apex about 144 px)
Air jump	1 (or unlimited in Flappy), at x0.88 power	
Coyote time	0.09 s	Can still jump just after leaving a ledge
Jump buffer	0.12 s	A jump pressed slightly early still works
Max fall speed	1300 px/s	
Player size	36 px (half-size 18) times scale	
Island floor	y = 400, 6 tiles of 100 px (x 180 to 780)	Classic layout
Classic platforms	(50,290,190), (720,290,190), (385,195,190)	x, y, width
Kill zone	y above 630, x below -90 or above 1050	
Shove cooldown	0.85 s	
Shove range	62 px x scale, plus the target's half size	
Shove force	(620 + 70 x hits taken) x sqrt(scale ratio)	Mega: x2.6
Shove pop-up	upward 260 px/s extra	
Stun after a hit	0.22 s	
Damage display	hits x 12 percent	Display only
Bomb fuse	4 + 0.6 x players alive, seconds	8 players: 8.8 s

Constant	Value	Meaning
Bomb holder speed	x1.12	
Bomb pass cooldown	0.7 s	
Bomb blast	radius 260 px, speed up to 760 px/s	
Crown holder speed	x0.9	
Crown steal protection	1.0 s	The player who lost it cannot instantly retake it
Crown respawn	1.5 s delay, 1.8 s shield	
Crate spawn	first at 4 s, then every 6 to 9 s, max 2, life 14 s	
Round time	Brawl 60 s, Bomb 75 s, Crown 45 s	
Twists per round	rounds 1 to 5: 1, 1, 2, 2, 3	
Time scale	final knockout 0.25 recovering at 1.5 per second	Visual feel

13.5 Network protocol sketch

```
-> join        { room: "KPQ7X", name: "Satya", color: 0, playerId: "uuid", token: "signed" }
<- joined      { you: 3, settings: {...}, players: [...], hostId: 1 }
-> settings:update { match: { rounds: 7 } }            (host only)
<- settings    { ...validated full settings... }
-> ready       { value: true }
-> start       {}                                      (host only)
<- round:start { n: 2, mode: "bomb", twists: ["ice"], theme: "neon", seed: 913742, slot: 5, pickedBy: 4 }
-> input       { seq: 1021, bits: 0b0101 }             (left, right, jump, shove)
<- state       { t: 5231, players: [[id,x,y,vx,vy,flags]...], bomb: {holder:3, fuse:4.2}, items: [...] }
<- event       { type: "bomb_pass", from: 3, to: 6 }
<- round:end   { ranking: [...], points: [...] }
<- match:end   { standings: [...] }
```

13.6 Screenshot index

File	What it shows	Real or mockup
`01-menu.png`	Menu	Real (v2 game code)
`02-intro-picked.png`	Round intro, picked twist	Real
`03-pick-twist.png`	Last place picks	Real
`04-play-brawl.png`	Last One Standing	Real
`05-play-bomb.png`	Bomb Tag	Real
`06-play-crown.png`	Crown Rush	Real
`07-play-lights-out.png`	Lights Out	Real
`08-play-crumble.png`	Crumbling Floor	Real
`09-results.png`	Round results	Real
`10-final.png`	Final standings	Real
`11-mock-landing.png`	Name prompt and join	MOCKUP
`12-mock-create-room.png`	Host settings	MOCKUP

File	What it shows	Real or mockup
`13-mock-lobby.png`	Lobby	MOCKUP
`14-mock-themes.png`	Theme picker	MOCKUP
`15-fig-arenas-rounds.png`	Six random arenas	Real output of the reference generator
`16-fig-arenas-settings.png`	Settings effect	Real output of the reference generator
`17-fig-power-spots.png`	First-touch race	Diagram on a generated arena
`18-fig-ad-placement.png`	Ad wireframes	Wireframe
`19-diagram-journey.png`	Player journey	Diagram
`20-diagram-architecture.png`	Architecture	Diagram

The real screenshots were rendered from the actual demo code with fixed random seeds, using the Bagel Fat One and Nunito fonts and the Noto Color Emoji font. In a phone browser the emoji will look different (section 8.3).

13.7 Source files

File	Description
`mutator-party.html`	v1 demo (one mode, 13 twists).
`mutator-party-v2.html`	v2 demo (3 modes, 4 items, last place picks).
Hosted demos	v1: https://claude.ai/artifact/VNNpjVsG7UR5UCFsR7ySup v2: https://claude.ai/artifact/Dx6moTfypwFB1oSgSpv5Qd

13.8 Test plan

Test	How	Pass condition
Round length	Headless simulation, 30 rounds per mode with bots	Brawl 40 to 60 s average, Bomb 45 to 75 s
Arena validity	Generate 10,000 arenas per setting set	100 percent reachable, no fallback needed
Determinism	Same seed on server and two clients	Identical platform lists
Settings abuse	Send out-of-range and wrong-type values	Server clamps, never crashes
Network	Throttle to 3G and 150 ms ping, drop packets 5 percent	Playable, no desync
Reconnect	Kill the connection mid-round	Back in under 30 s, bot covers the gap
Host leave	Host closes the tab in lobby and mid-match	New host chosen, match continues
Load	100 simulated rooms of 8 bots	CPU under 70 percent on the target server
Phones	3 real phones (one low-end Android, one iPhone, one tablet)	60 fps, buttons usable, no scroll
Accessibility	Reduce motion on, colors simulated for color blindness	Still playable and readable
Ads	Test ads in staging	Never visible during a round

13.9 Open questions for you

1. Should the first public version be **English only**, or Hinglish text from the start?
2. Do you want **accounts** early (cosmetics, friends list) or a strictly guest-only first launch?
3. Which first audience: **college students** (fest and hostel rooms) or **general casual players**?
4. Is the target a **web game first** (recommended) or an **Android app** (Flutter) first?
5. Will you keep the name **Mutator Party**? A trademark check should come before more branding work.

End of document.